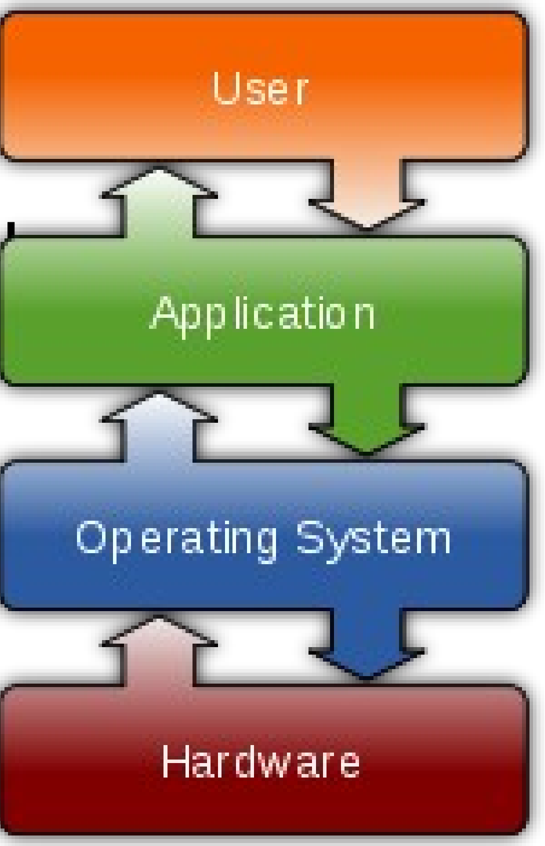
3. Hafta

İŞLETİM SİSTEMLERİ VE LINUX (KALİ LİNUX)

İşletim Sistemi

Operating **S**ystem

Donanımın doğrudan denetimi ve yönetiminden, temel sistem işlemlerinden ve uygulama programlarını çalıştırmaktan sorumlu olan sistem yazılımıdır.

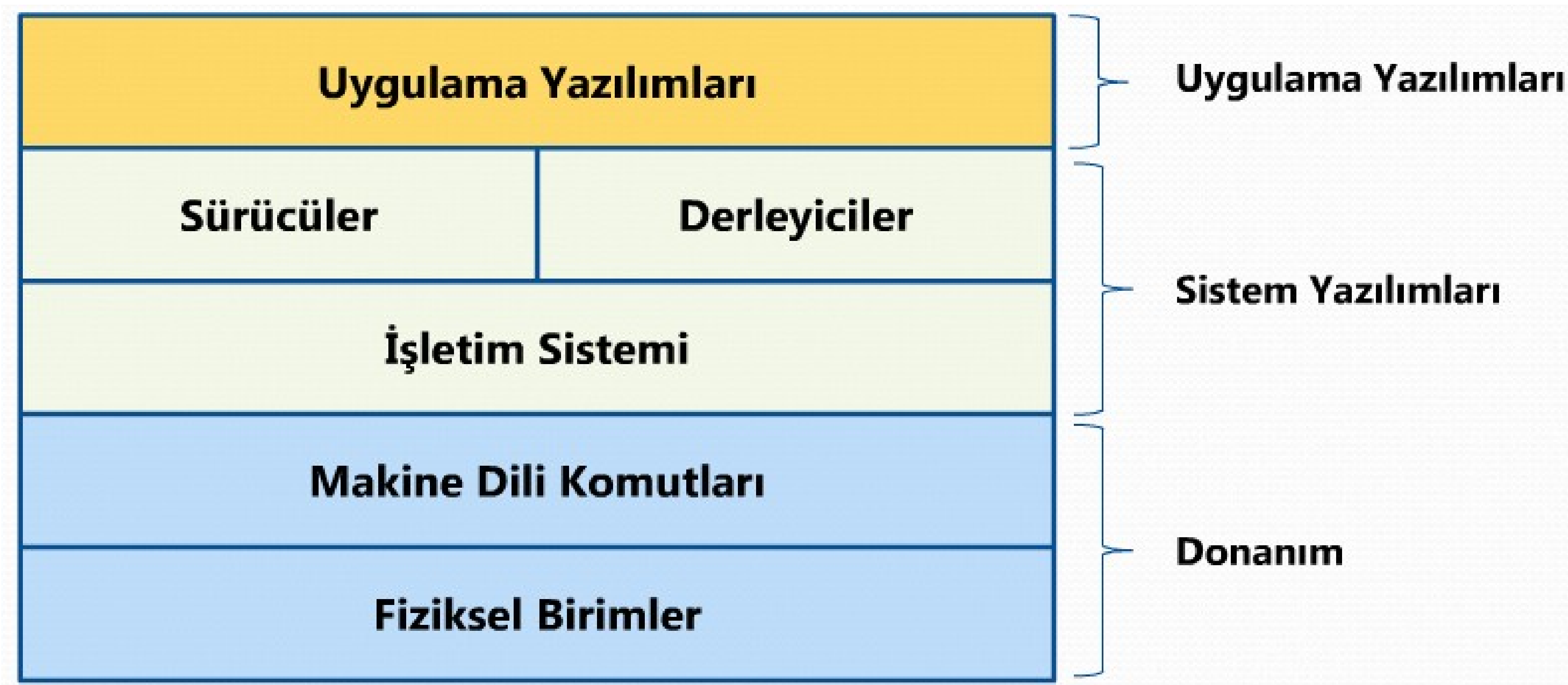


Donanım <-> Çekirdek <-> Kabuk <-> Uygulamalar

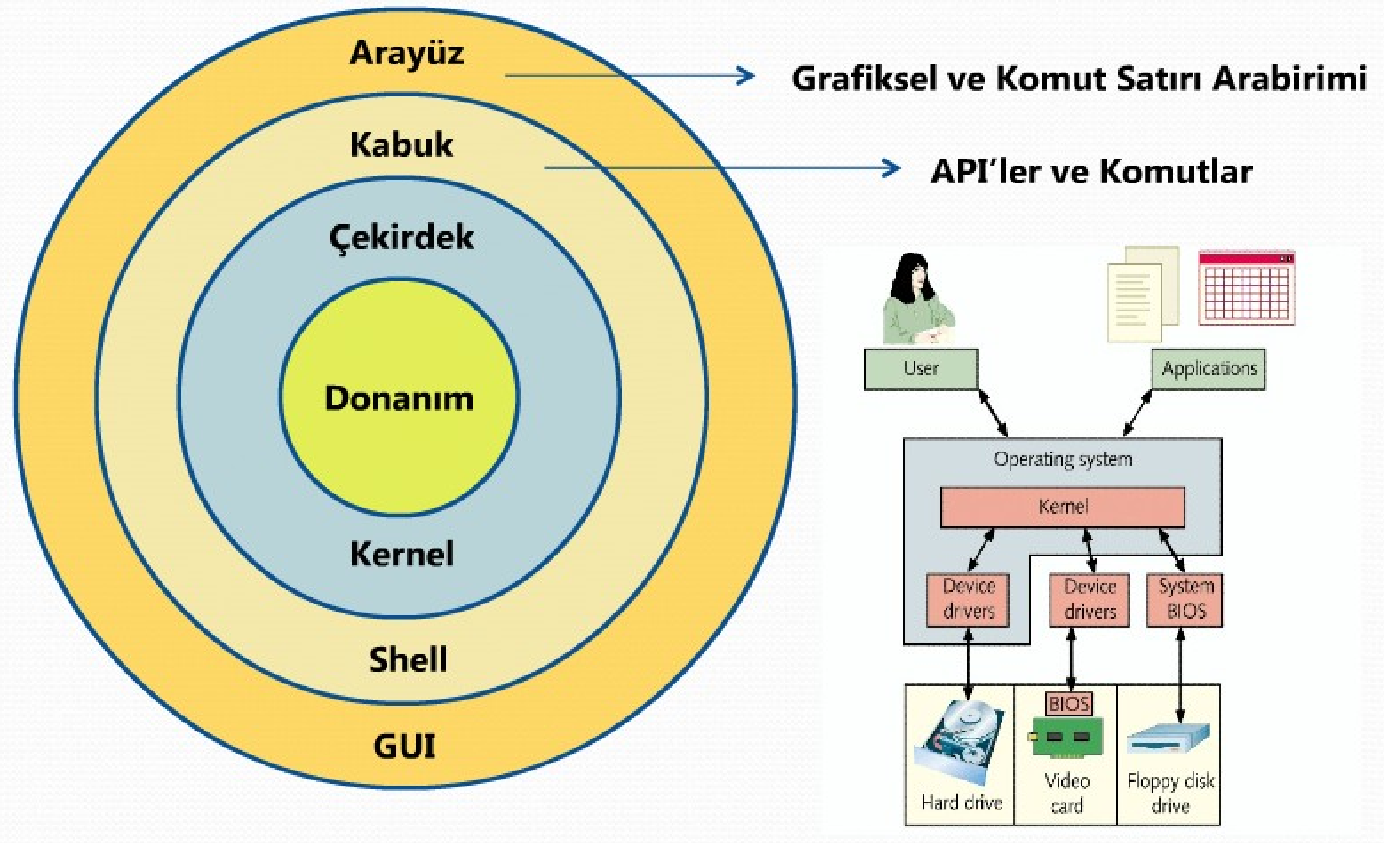
İşletim Sistemi

Donanım ile kullanıcı arasında bir arayüz (interface) görevini, aynı zamanda donanım ile yazılım birimlerinin etkin bir biçimde kullanılmalarını sağlayan sistem programları topluluğudur.

İşletim Sistemi Mimarileri



İşletim Sistemi Mimarileri



İşletim Sistemi Türleri

- Kişisel bilgisayar(PC),
- Sunucu (Server),
- Anaçatı (Mainframe),
- Çok işlemcili (Parallel computing),
- Gerçek zamanlı (Real-time),
- Gömülü (Embedded),

işletim sistemleri..

İşletim Sistemi Çeşitleri

- Unix ve çeşitleri
- Linux ve dağıtımları
- Windows ve sürümleri
- Apple OS ve IOS sürümleri
- Android ve sürümleri

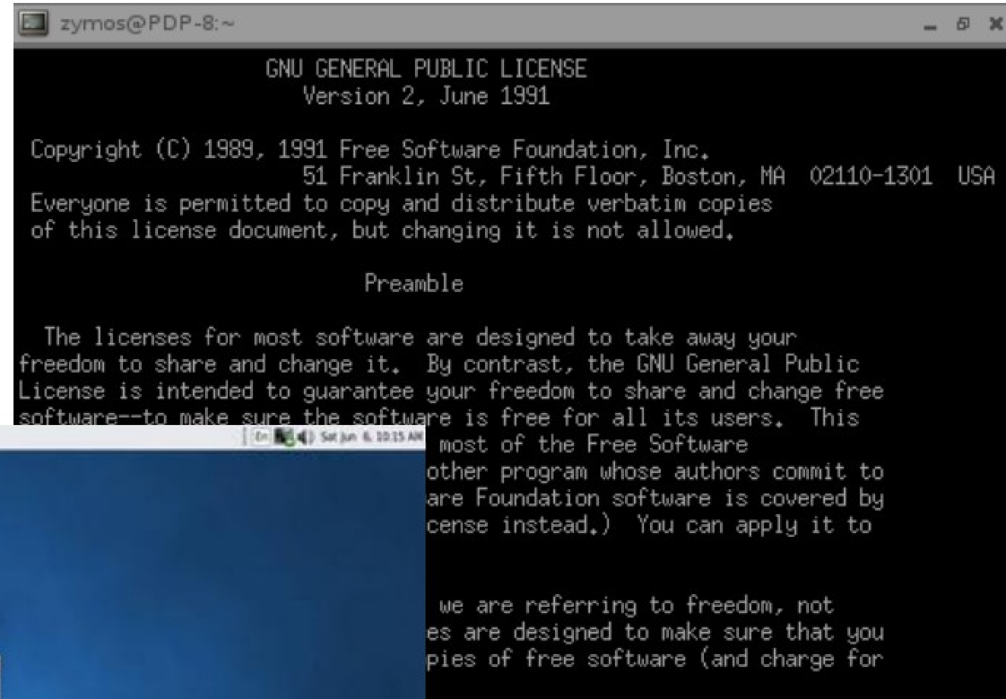
Unix

- 1969’da Ken Thompson ve Dennis Ritchie katılımı ile AT&T Bell Laboratuvarları’nda geliştirildi.
- Geliştirme süreci sonunda **UNIX** adını aldı
- MULTICS’in versiyonu olan PDP-7 mini bilgisayarı üzerinde UNICS’i yazdı.
- DEC PDP-7’lerde 8K bellekler ile çalıştırıldı.
- İlk olarak Assembly dilinde yazıldı.

Unix

- 60’lı yılların sonunda “C” diliyle yazılmıştır.
- Çok kullanıcılı *(multiuser)* ve aynı anda birçok işi yapabilen *(multitasking)* bir işletim sistemidir.
- Komut yorumlayıcı programlar *(shell)* aracılığı ile kullanıcı ve bilgisayar sisteminin iletişimi sağlanır.
- 1980 ler de Unix ~ fiyatı 1300-1850$..
- Pek çok Unix çeşidi vardır.
  - BSD Unix, OpenSolaris, HP-UX, AIX, SCO Unix, Sun OS…

Unix

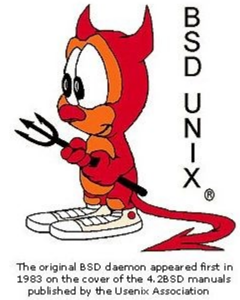


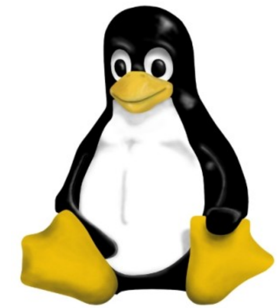
Grafik Kullanıcı Arayüzü

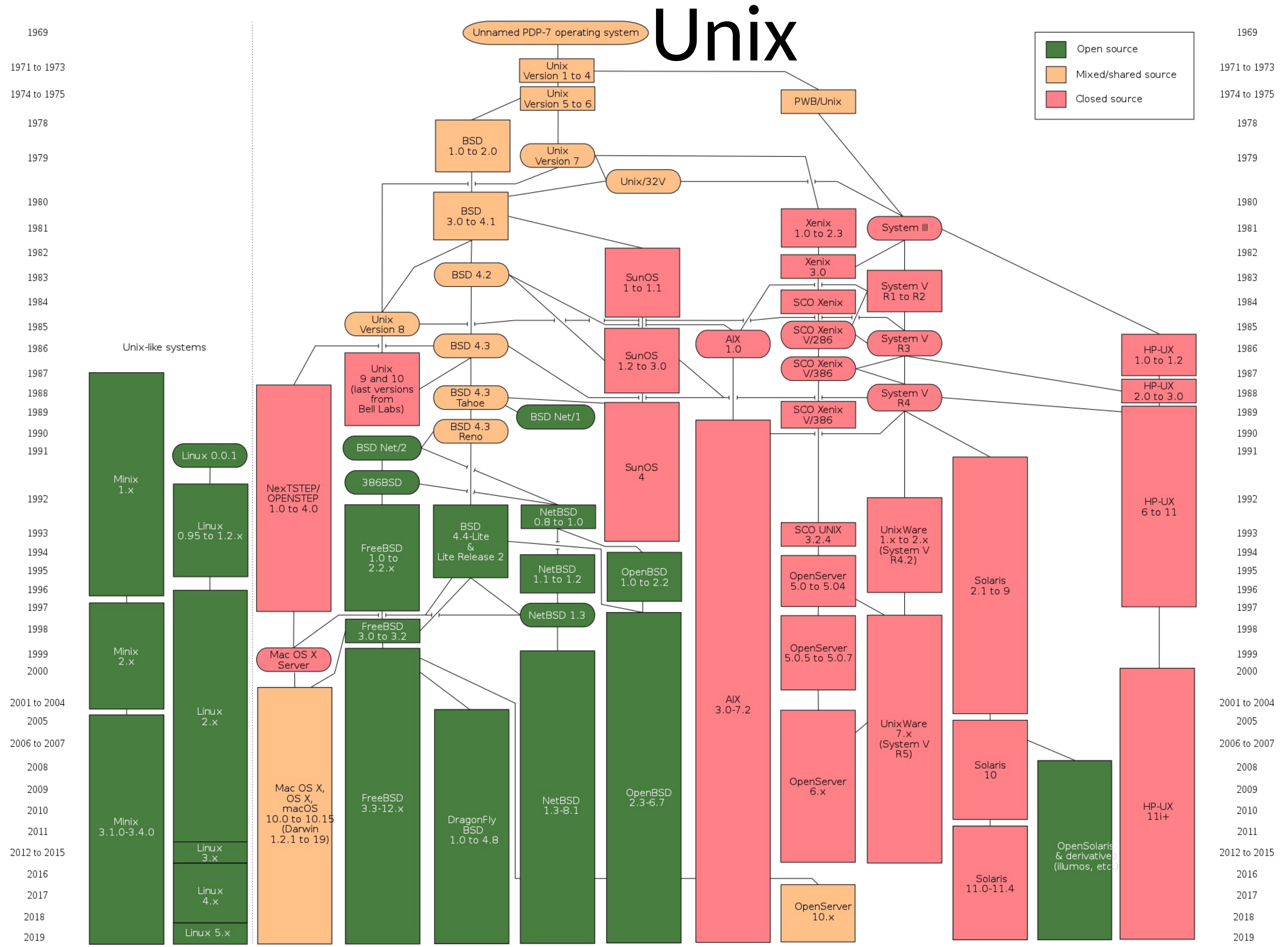
Metin Tabanlı Arayüz

Unix

- BSD Unix
- Solaris
- OpenSolaris
- Linux
- HP-UX
- AIX
- Minix
- SCO Unix
- Sun OS
- Digital

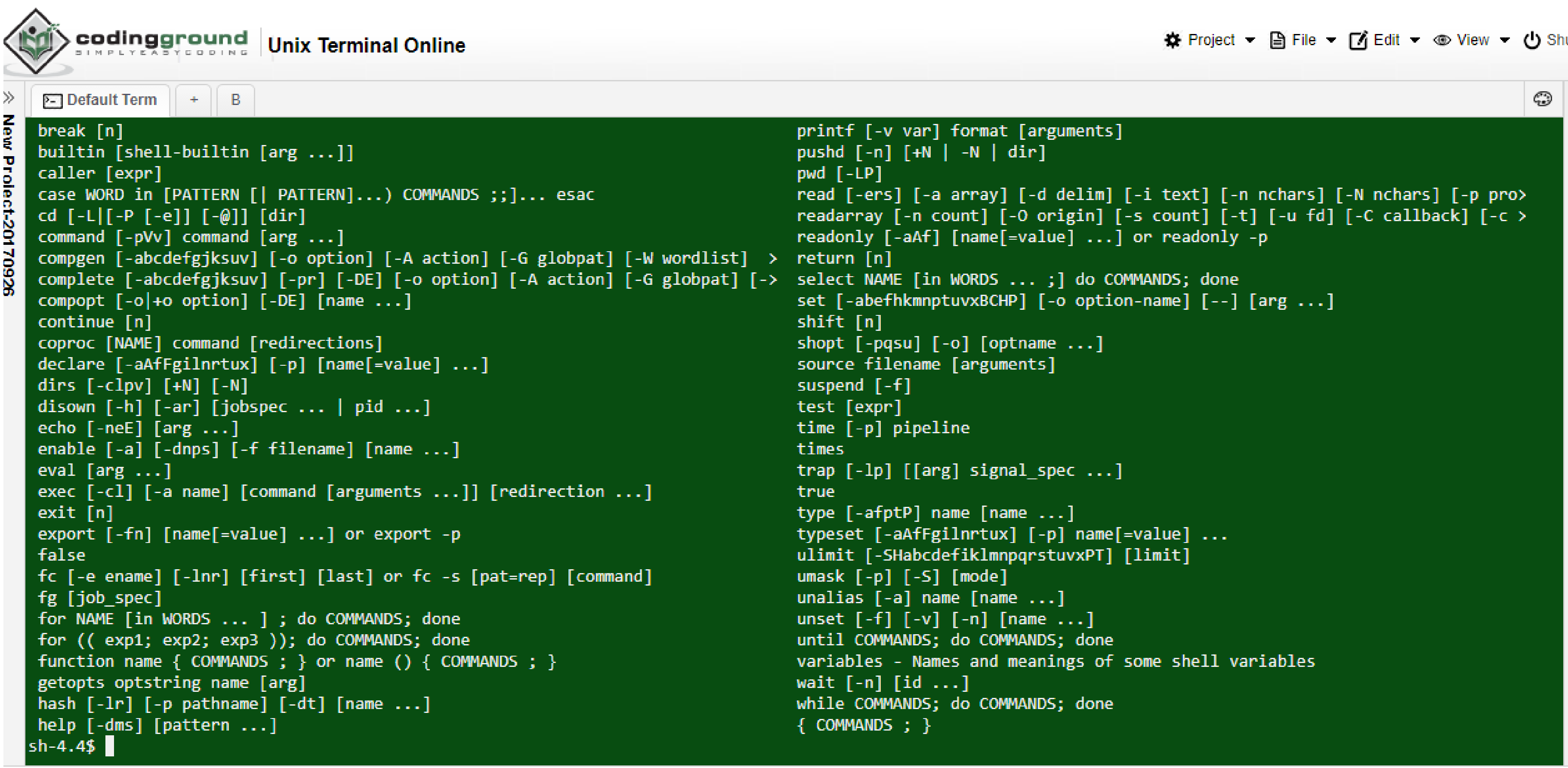





Unix
Open source
Mixed/shared source
Closed source
Unix-like systems
1969
1971 to 1973
1974 to 1975
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001 to 2004
2005
2006 to 2007
2008
2009
2010
2011
2012 to 2015
2016
2017
2018
2019
Unnamed PDP-7 operating system
Unix Version 1 to 4
Unix Version 5 to 6
PWB/Unix
BSD 1.0 to 2.0
Unix Version 7
Unix/32V
BSD 3.0 to 4.1
Xenix 1.0 to 2.3
System III
Xenix 3.0
SunOS 1 to 1.1
BSD 4.2
System V R1 to R2
SCO Xenix
Unix Version 8
AIX 1.0
SCO Xenix V/286
System V R3
BSD 4.3
SunOS 1.2 to 3.0
HP-UX 1.0 to 1.2
Unix 9 and 10 (last versions from Bell Labs)
SCO Xenix V/386
BSD 4.3 Tahoe
System V R4
HP-UX 2.0 to 3.0
BSD Net/1
SCO Xenix V/386
BSD 4.3 Reno
BSD Net/2
Linux 0.0.1
Minix 1.x
NexTSTEP/ OPENSTEP 1.0 to 4.0
SunOS 4
386BSD
NetBSD 0.8 to 1.0
HP-UX 6 to 11
Linux 0.95 to 1.2.x
FreeBSD 1.0 to 2.2.x
BSD 4.4-Lite & Lite Release 2
SCO UNIX 3.2.4
UnixWare 1.x to 2.x (System V R4.2)
NetBSD 1.1 to 1.2
OpenBSD 1.0 to 2.2
OpenServer 5.0 to 5.04
Solaris 2.1 to 9
NetBSD 1.3
FreeBSD 3.0 to 3.2
Minix 2.x
Mac OS X Server
OpenServer 5.0.5 to 5.0.7
Linux 2.x
AIX 3.0-7.2
UnixWare 7.x (System V R5)
Solaris 10
Minix 3.1.0-3.4.0
Mac OS X, OS X, macOS 10.0 to 10.15 (Darwin 1.2.1 to 19)
FreeBSD 3.3-12.x
DragonFly BSD 1.0 to 4.8
NetBSD 1.3-8.1
OpenBSD 2.3-6.7
OpenServer 6.x
HP-UX 11i+
Linux 3.x
OpenSolaris & derivatives (illumos, etc.)
Linux 4.x
Solaris 11.0-11.4
OpenServer 10.x
Linux 5.x

Online Unix Terminal

http://www.tutorialspoint.com/unix_terminal_online.php



GNU - GPL

- Richard M. Stallman yazılımların koşullarını kabullenmek istemiyor..
- 1984 GNU projesini başlatıyor..
- GNU = **G**NU is **N**ot **U**nix
- 1985'de Free Software Foundation (Özgür Yazılım Vakfı) kuruluyor..
- 1991'de **G**eneral **P**ublic **L**icence (Genel Kamu Lisansı)
- Özgür ve açık kaynak kodlu..
- Kaynak kodlar üzerinde herkes istediği değişikliği yapabilir, dağıtabilir, satabilir.
- Yapılan değişikliğinde kodları paylaşılmalı..

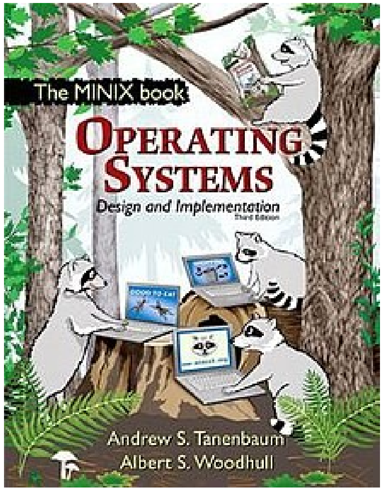
FREE SOFTWARE
FOUNDATION

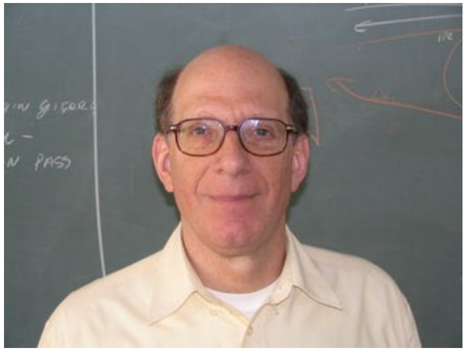
GPLv3 Free Software
Free as in Freedom

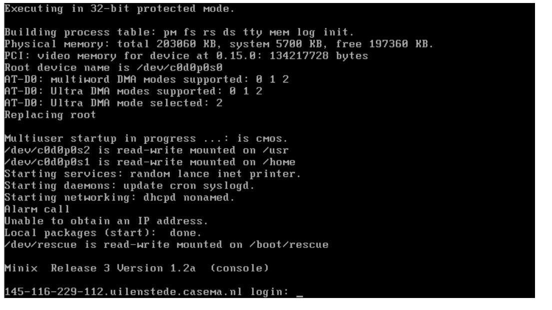
https://www.youtube.com/watch?v=_St4bC9fcfA **Richard Stallman Özgür Yazılım TEDx Konuşması** (2014)

Minix

- Dr.Andrew Stuart "Andy" Tanenbaum, Minix işletim sistemini geliştirmiştir.
- Öğrencilerine Unix yerine Minix üzerinde eğitim-uygulama yaptırmıştır.
- İşletim Sistemleri (Minix) kitabını yazmıştır.







Minix


MINIX 1.5 5¼'' for the IBM® PC, XT,™ AT,® 386 and PS/2®
by Andrew S. Tanenbaum
© 1991 by Prentice-Hall, Inc., A Division of Simon & Schuster
ISBN 0-13-585076-2
10 9 8 7 6 5 4 3 2 1
DISK 1 PC Boot Disk

Minix

```
Executing in 32-bit protected mode.

Building process table: pm fs rs ds tty mem log init.
Physical memory: total 203060 KB, system 5700 KB, free 197360 KB.
PCI: video memory for device at 0.15.0: 134217728 bytes
Root device name is /dev/c0d0p0s0
AT-D0: multiword DMA modes supported: 0 1 2
AT-D0: Ultra DMA modes supported: 0 1 2
AT-D0: Ultra DMA mode selected: 2
Replacing root

Multiuser startup in progress ...: is cmos.
/dev/c0d0p0s2 is read-write mounted on /usr
/dev/c0d0p0s1 is read-write mounted on /home
Starting services: random lance inet printer.
Starting daemons: update cron syslogd.
Starting networking: dhcpd nonamed.
Alarm call
Unable to obtain an IP address.
Local packages (start):  done.
/dev/rescue is read-write mounted on /boot/rescue

Minix  Release 3 Version 1.2a  (console)

145-116-229-112.uilenstede.casema.nl login: _
```

Linux

- Helsinki Üniversitesi'nde 23 yaşında, Finlandiyalı bir öğrenci,
- **Linus Torvalds,**
- Minix'ten esinlenerek Linux işletim sistemini (çekirdeği) oluşturmuştur..
- 5 Ekim 1991 – Linux 0.02 internet ve haber gruplarında yer alıyor..
- GNU-GPL Lisansı ile dağılıyor..

https://www.youtube.com/watch?v=o8NPllzkFhE
Linus Torvalds TED Konuşması (2016) (Türkçe Altyazılı)

Linux

Hello everybody out there using minix -

I'm doing a (free) operating system (just a hobby, won't be big and professional like gnu) for 386(486) AT clones. This has been brewing since april, and is starting to get ready. I'd like any feedback on things people like/dislike in minix, as my OS resembles it somewhat (same physical layout of the file-system (due to practical reasons) among other things).

I've currently ported bash(1.08) and gcc(1.40), and things seem to work. This implies that I'll get something practical within a few months, and I'd like to know what features most people would want. Any suggestions are welcome, but I won't promise I'll implement them :-)

Linus (torvalds@kruuna.helsinki.fi)
PS. Yes – it's free of any minix code, and it has a multi-threaded fs. It is NOT portable (uses 386 task switching etc), and it probably never will support anything other than AT-harddisks, as that's all I have :-(.
—Linus Torvalds

Linus Torvalds'ın 5 Ağustos 1991'de, comp.os.minix adresli, haber grubuna gönderdiği mesaj.

17 Eylül 1991'de Linux'un ilk sürümü olan 0.01'i İnternet'te yayınladı.

5 Ekim 1991'de temel özellikleriyle beraber ilk resmi Linux sürümü 0.02'yi yayınladı.

rom: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)
lewsgroups: comp.os.minix
Subject: What would you like to see most in minix?
Summary: small poll for my new operating system
Message-ID: <1991Aug25.205708.9541@klaava.Helsinki.FI>
Date: 25 Aug 91 20:57:08 GMT
Organization: University of Helsinki

Hello everybody out there using minix -
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professional like gnu) for 386(486) AT clones. This has been brewing
since april, and is starting to get ready.I'd like any feedback on
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—Linus Torvalds

Andrew Tanenbaum's f...

www.linfo.org/obsolete.html

Linux Today | Practical Technolo... | Computerworld | Slashdot | Twitterfall | Other Bookmarks

LINFO

"LINUX is obsolete"

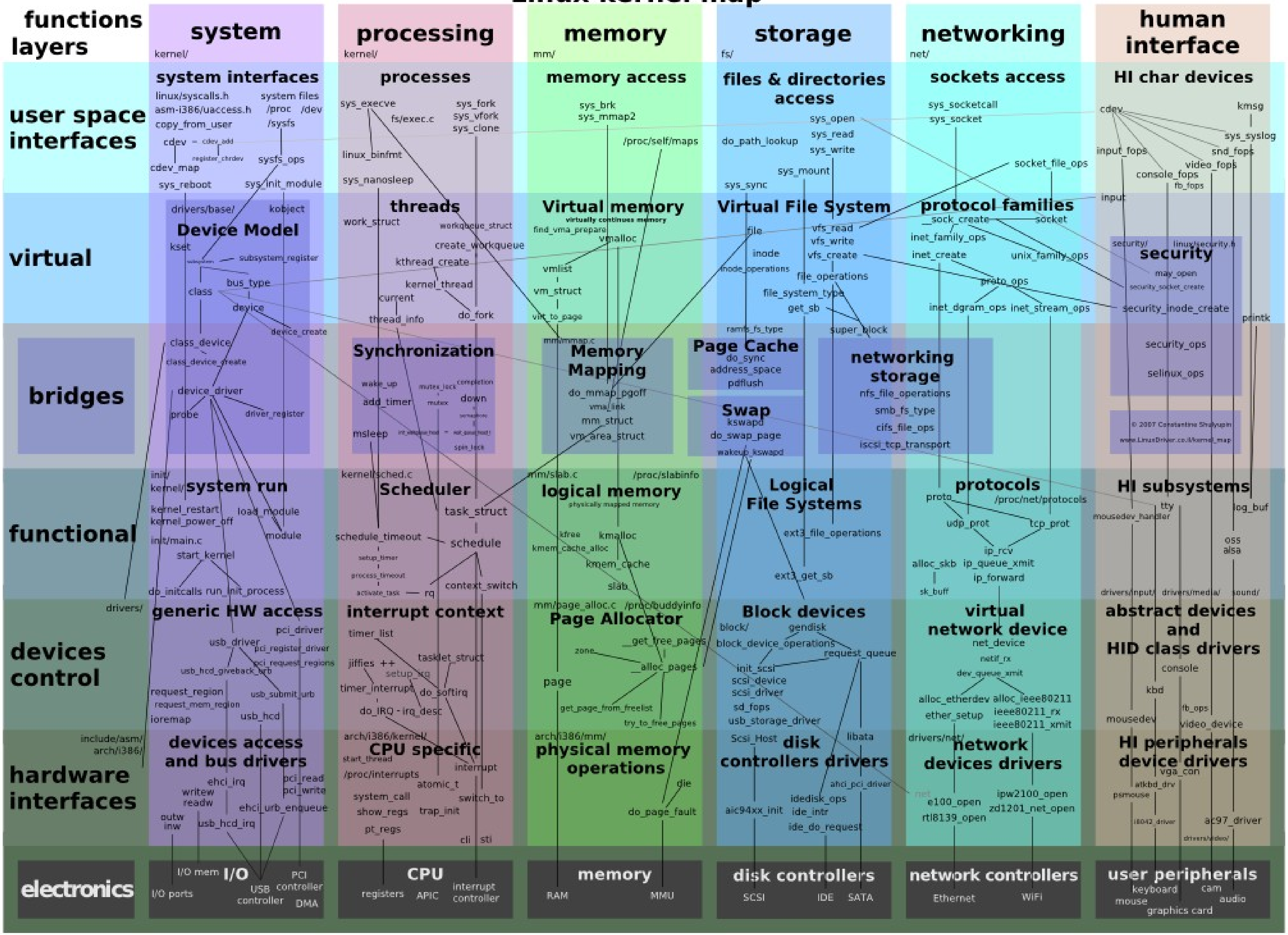
Below is Andrew Tanenbaum's famous 1992 comp.os.minix newsgroup posting in which he criticizes Linux's monolithic kernel as well as Linus Torvald's first response in a very long thread.

```
From: ast@cs.vu.nl (Andy Tanenbaum)
Newsgroups: comp.os.minix
Subject: LINUX is obsolete
Date: 29 Jan 92 12:12:50 GMT
Organization: Fac. Wiskunde & Informatica, Vrije Universiteit, Amsterdam


I was in the U.S. for a couple of weeks, so I haven't commented much on
LINUX (not that I would have said much had I been around), but for what
it is worth, I have a couple of comments now.

As most of you know, for me MINIX is a hobby, something that I do in the
evening when I get bored writing books and there are no major wars,
revolutions, or senate hearings being televised live on CNN.  My real
job is a professor and researcher in the area of operating systems.

As a result of my occupation, I think I know a bit about where operating
are going in the next decade or so.  Two aspects stand out:

1. MICROKERNEL VS MONOLITHIC SYSTEM
   Most older operating systems are monolithic, that is, the whole operating
   system is a single a.out file that runs in 'kernel mode.'  This binary
   contains the process management, memory management, file system and the
   rest. Examples of such systems are UNIX, MS-DOS, VMS, MVS, OS/360,
   MULTICS, and many more.

   The alternative is a microkernel-based system, in which most of the OS
   runs as separate processes, mostly outside the kernel.  They communicate
   by message passing.  The kernel's job is to handle the message passing,
   interrupt handling, low-level process management, and possibly the I/O.
   Examples of this design are the RC4000, Amoeba, Chorus, Mach, and the
   not-yet-released Windows/NT.

   While I could go into a long story here about the relative merits of the
   two designs, suffice it to say that among the people who actually design
   operating systems, the debate is essentially over.  Microkernels have won.
   The only real argument for monolithic systems was performance, and there
   is now enough evidence showing that microkernel systems can be just as
   fast as monolithic systems (e.g., Rick Rashid has published papers
   comparing Mach 3.0 to monolithic systems) that it is now all over but
   the shoutin'.
```

Linux Çekirdeği (Kernel)

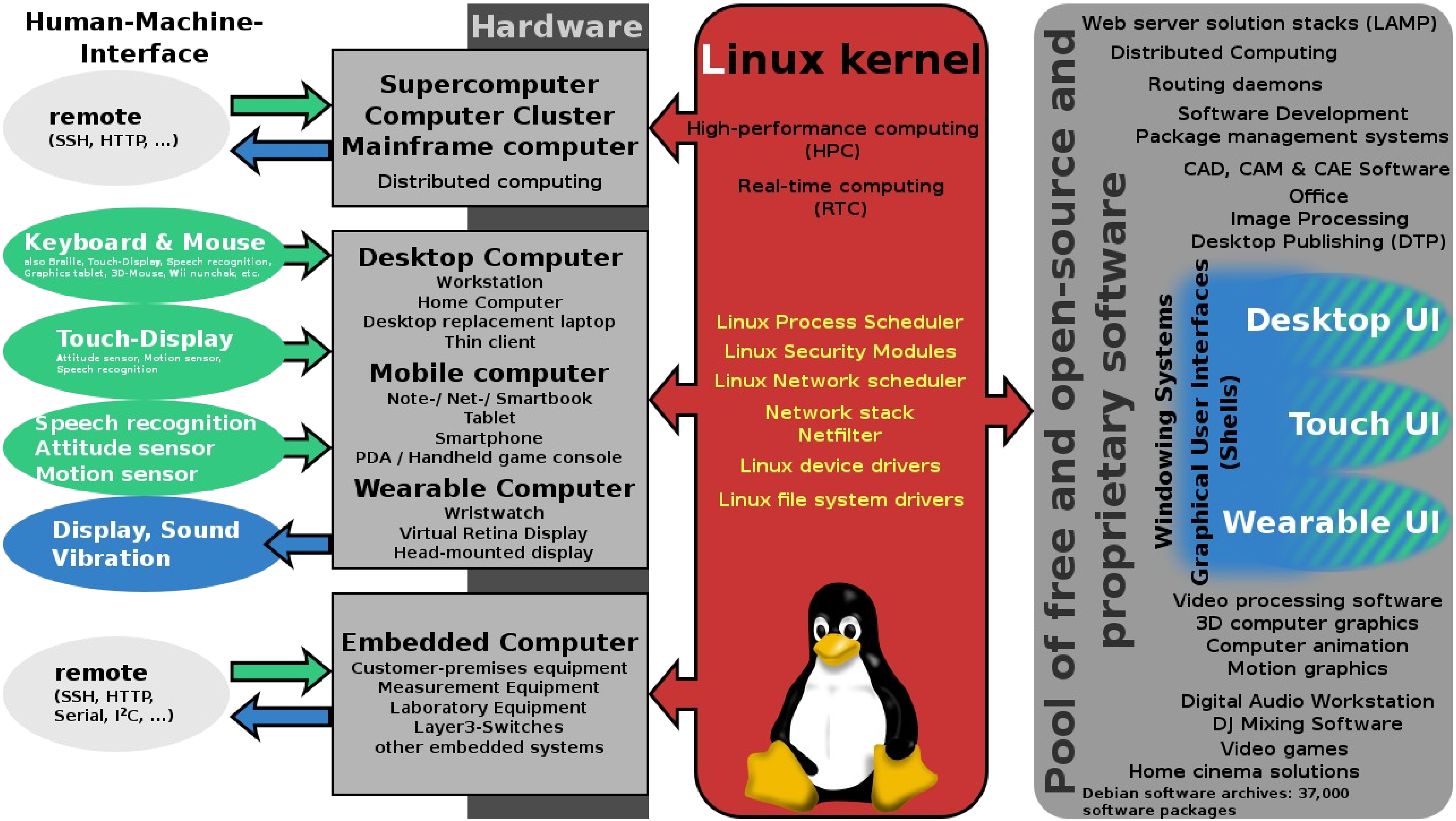
- Multitasking (Çok görevli)
- Virtual Memory (Sanal Bellek)
- Protected Mode (Korumalı Mod)
- Hızlı TCP/IP
- Çoklu kullanıcı ortamı (Multi User)
- Modüler Yapı
- İstenilen şekilde yapılandırma yeteneği
- Modern bir işletim sistemi çekirdeğinden beklenecek pek çok özellik ve daha da fazlası

Linux kernel map

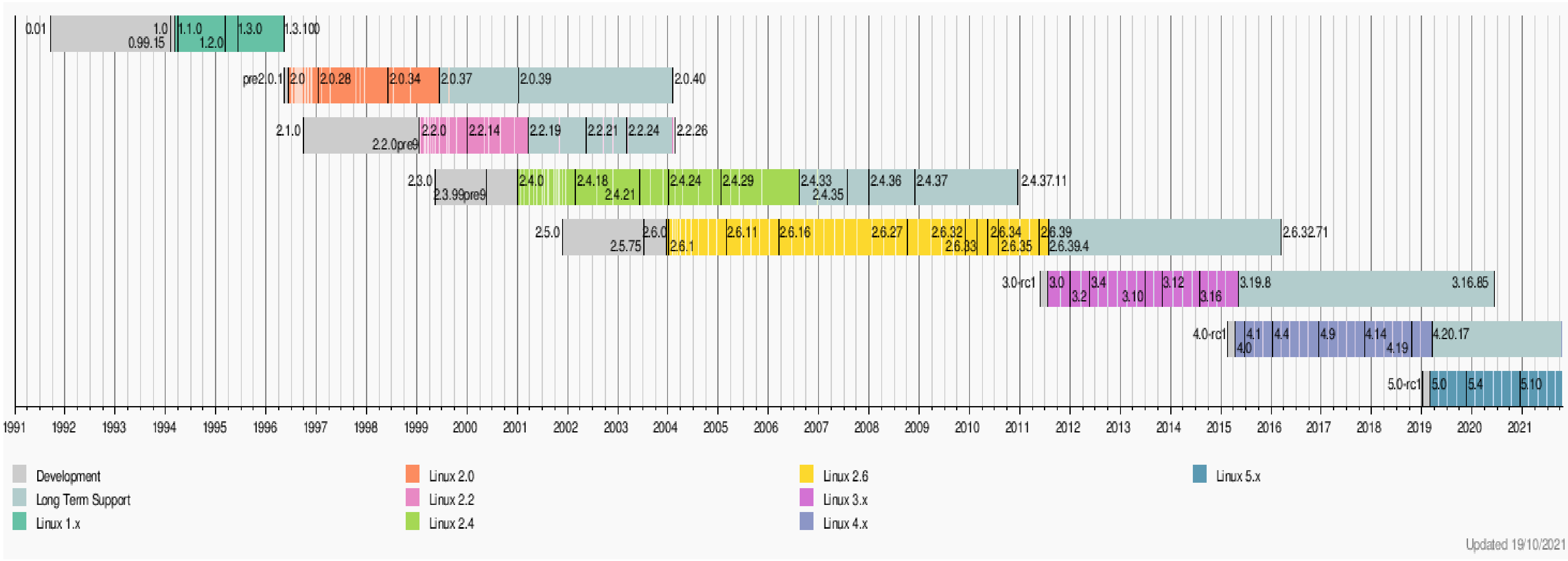


https://en.wikipedia.org/wiki/Linux_kernel

Linux Çekirdeği’nin Desteklediği Donanım ve Dahil Olduğu Mimariler



Linux Çekirdeği (Kernel) Zaman ve Sürüm Çizelgesi



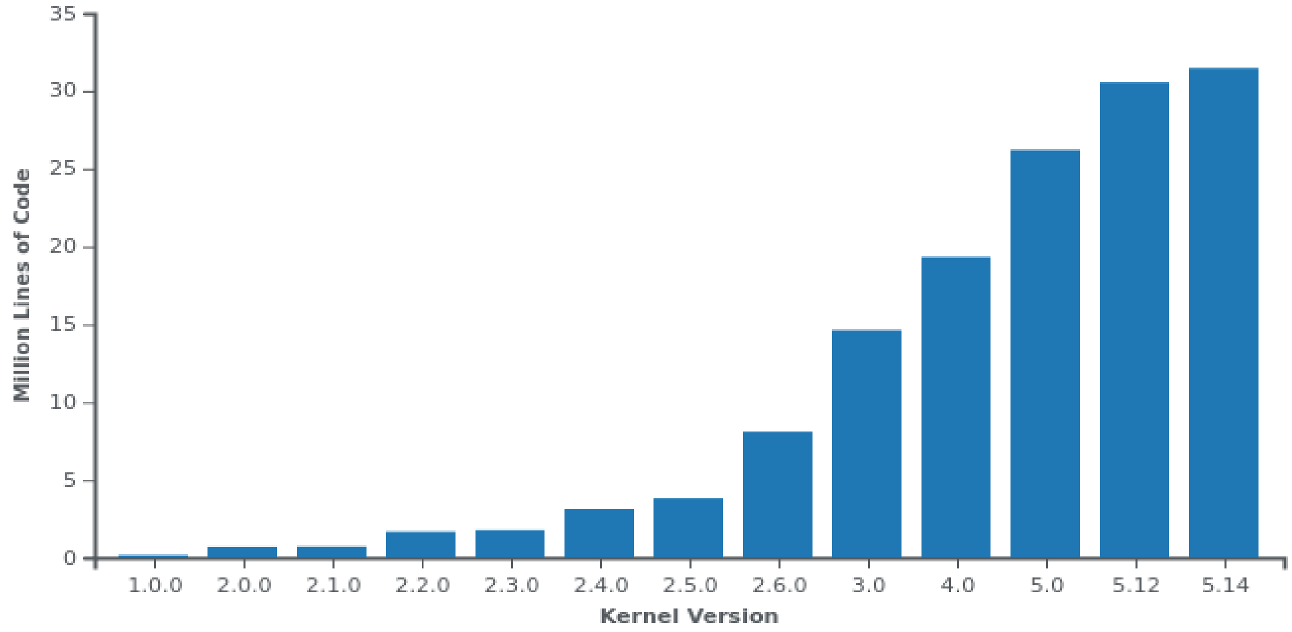
En Son Sürüm **5.15**

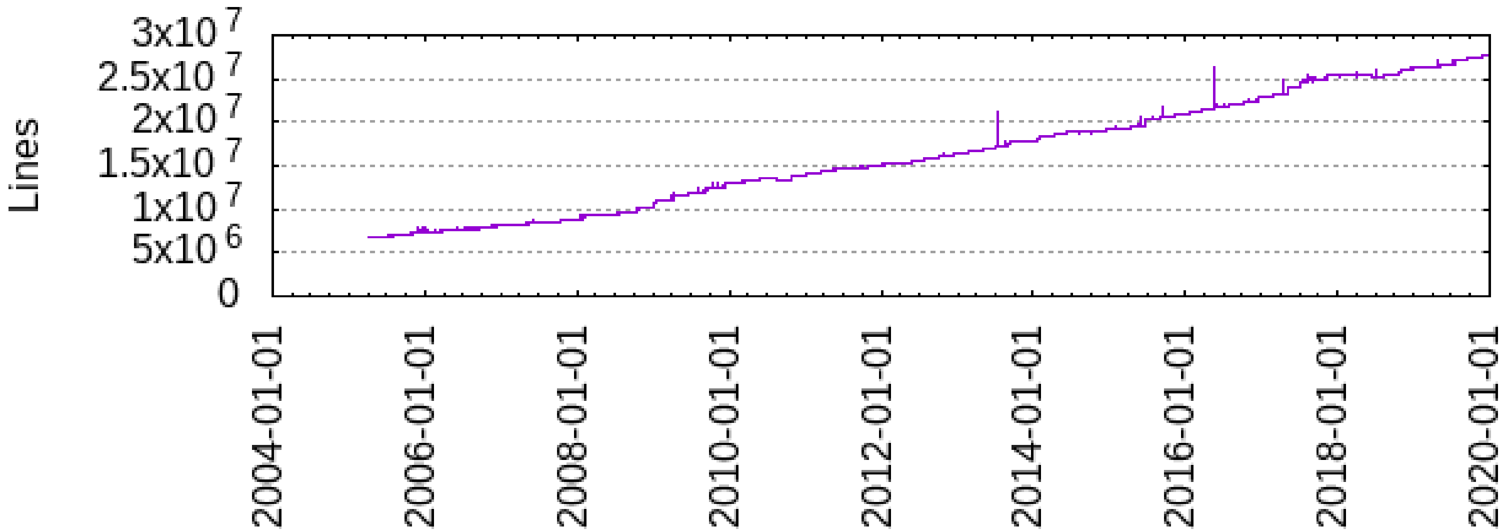
https://en.wikipedia.org/wiki/Linux_kernel_version_history

Ekim 2021

Linux Çekirdeği İstatistikleri

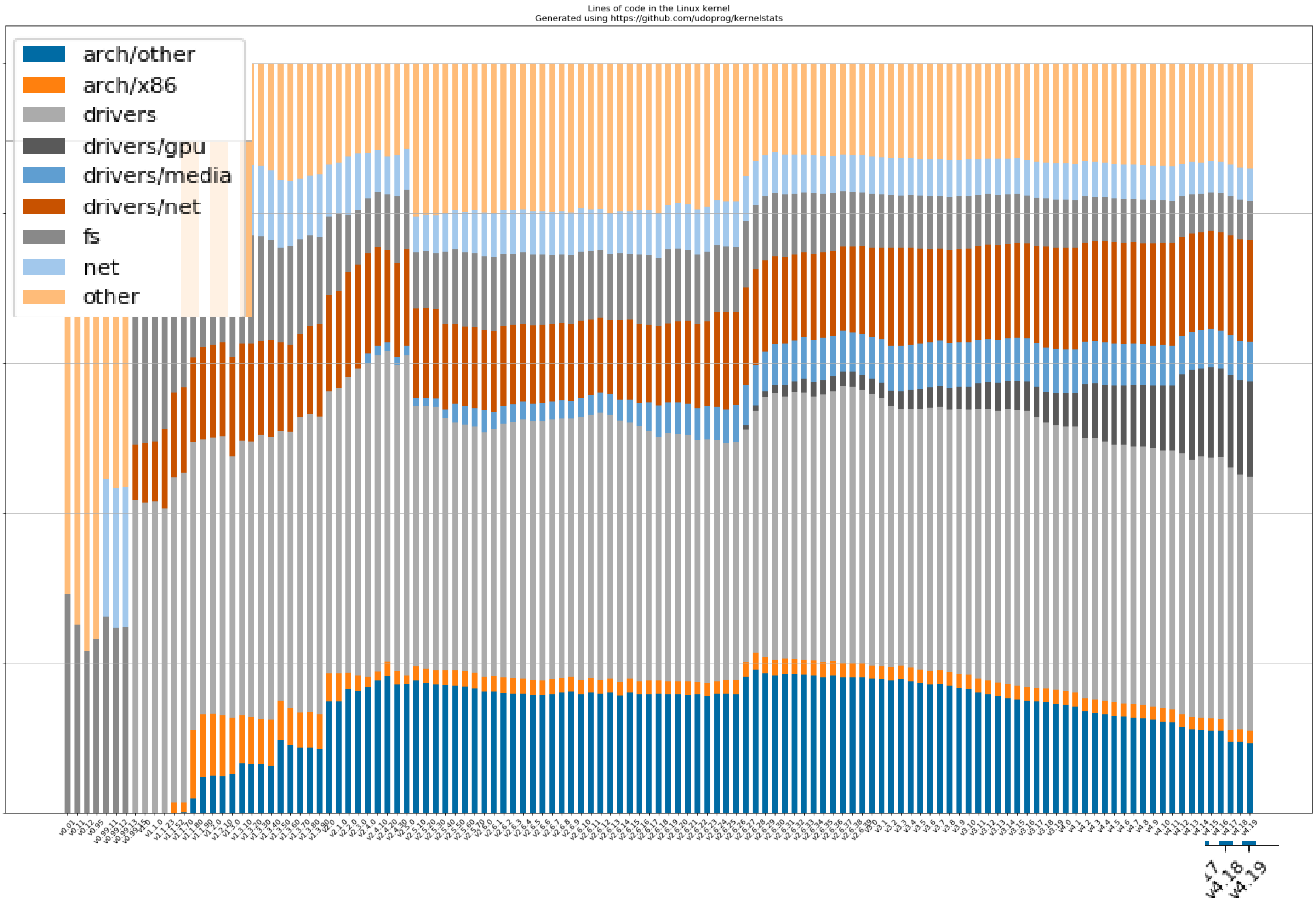
Satır Sayısı : **25.584.633**
Dosya Sayısı : **61.725**
Yazar Sayısı : **19.009**



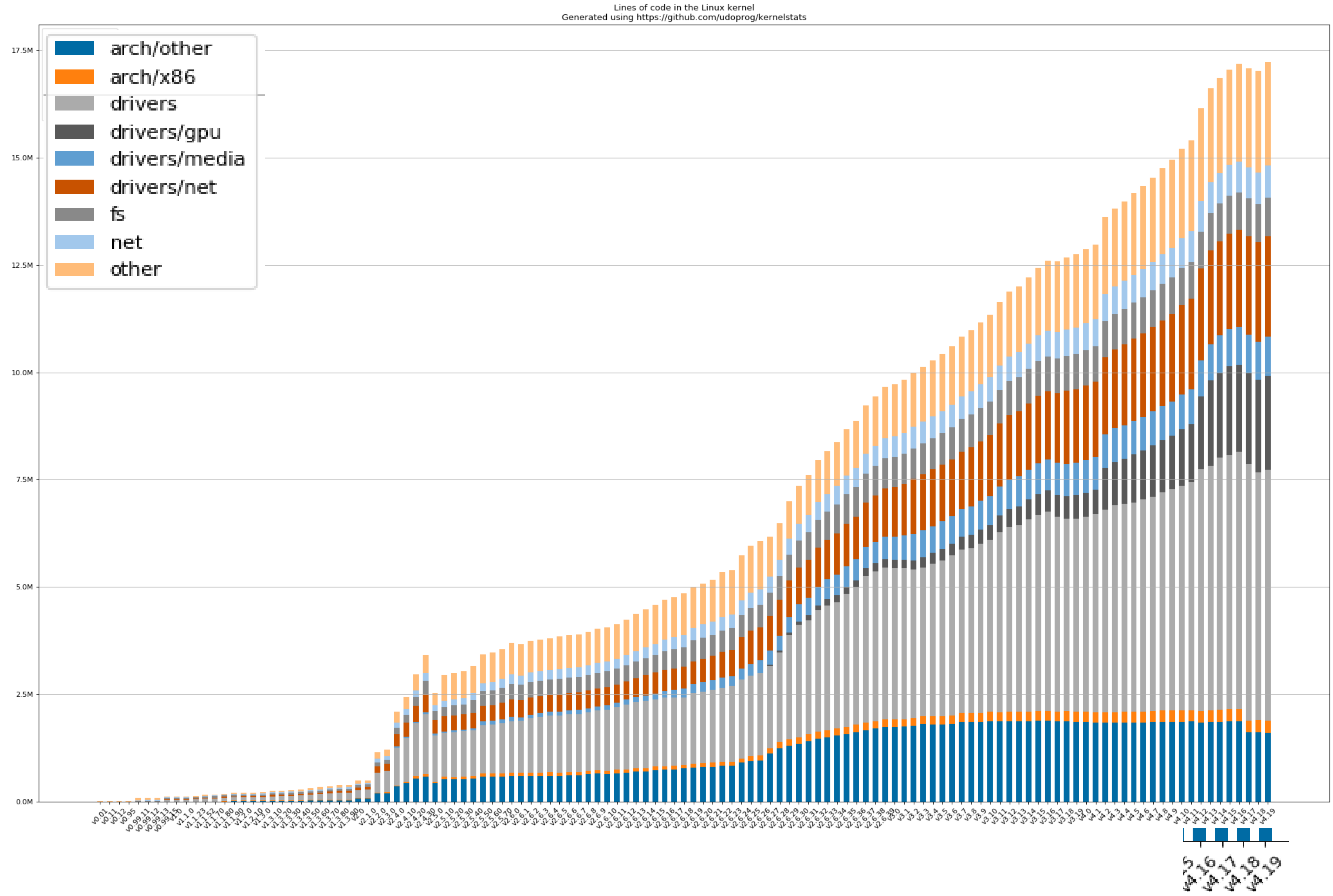


Veriler Ekim 2021 itibariyle..

Kaynaklar:
wikipedia.org/wiki/Linux_kernel
https://www.phoronix.com/scan.php?page=news_item&px=Linux-Git-Stats-EOY2019



github.com/udoprog/kernelstats



github.com/udoprog/kernelstats

Online Linux Terminal

Linux Terminal (Üyelik)

- http://www.webminal.org/terminal/

Linux Terminal (JavaScript)

- https://bellard.org/jslinux/

- Ubuntu Tour(Demo) web

http://tour.ubuntu.com/en/

Açık Kaynak (Open Source)

- Kaynak kodu isteyen herkese açık olan yazılımdır.
- Kullanıcıya değiştirme özgürlüğü sağlar.

open source initiative

copyleft

Özgür Yazılım (*Free software*)

Kullanıcısına;

- Çalıştırma,
- Kopyalama,
- Dağıtma,
- İnceleme,
- Değiştirme ve
- Geliştirme

özgürlükleri tanıyan **yazılım** türüdür.

Özgür Yazılım

- Herhangi bir amaç için yazılımı **çalıştırma** özgürlüğü (0)
- Her ne istiyorsanız onu yaptırmak için programın nasıl çalıştığını öğrenmek ve onu **değiştirme** özgürlüğü (1)

(Yazılımın kaynak koduna ulaşmak, bu iş için önkoşuldur.)

- Kopyaları **dağıtma** özgürlüğü. Böylece komşunuza yardım edebilirsiniz (2)
- Tüm toplumun yarar sağlayabileceği şekilde programı **geliştirme** ve geliştirdiklerinizi (ve genel olarak değiştirilmiş sürümlerini) **yayınlama** özgürlüğü (3).

(Kaynak koduna erişmek, bunun için bir önkoşuldur.)

- *Kaynak ve Ayrıntılar:* https://www.gnu.org/philosophy/free-sw.tr.html

Linux Dağıtımları

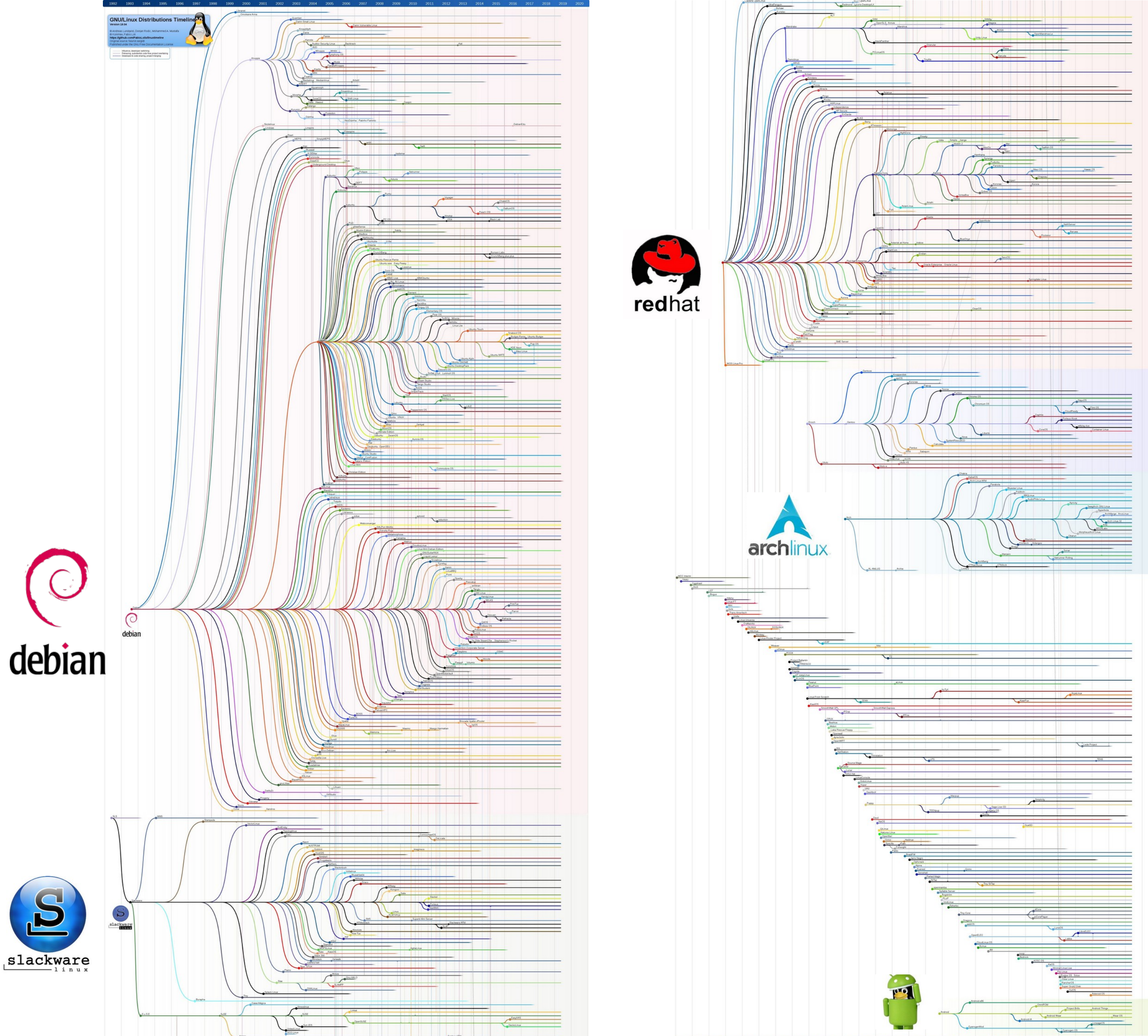
- Dağıtım, bir GNU/Linux sistemini kurmayı ve yönetmeyi kolaylaştırmayı amaçlayan yazılımlar bütünüdür.

KALI
BY OFFENSIVE SECURITY

KALI LINUX™

ubuntu debian fedora pardus KNOPPIX openSUSE

DistroWatch.com
Put the fun back into computing. Use Linux, BSD.


GNU/Linux Distributions Timeline
Version 19.04
© Andreas Lundqvist, Donjan Rodic, Mohammed A. Mustafa
© Konimex, Fabio Loli
https://github.com/FabioLolix/linuxtimeline
Original source: futurist.se/gldt
Published under the GNU Free Documentation License
Influence, developer switching
Rebasing, substantial code flow, project overtaking
Developer & code sharing, project merging
debian
redhat
archlinux
slackware linux

DistroWatch.com
Put the fun back into computing. Use Linux, BSD.

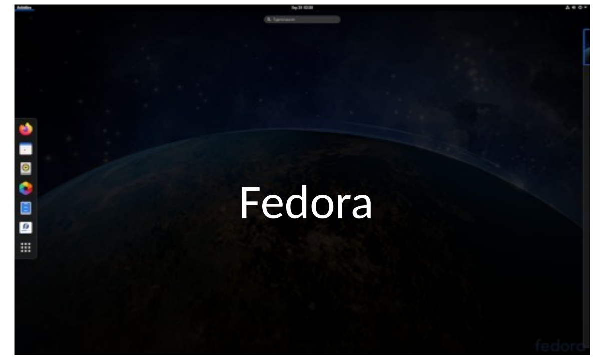
Dağıtımlar

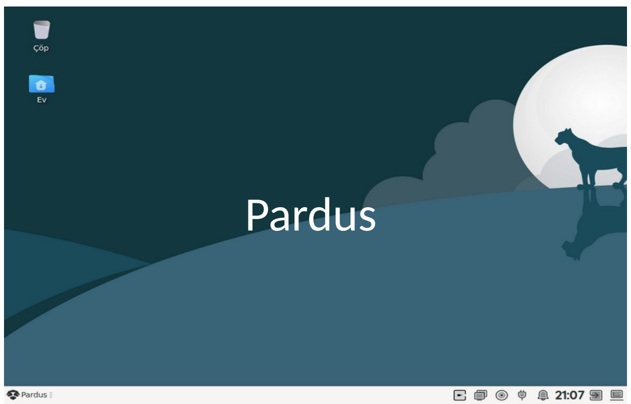
2001 yılından beri tüm dağıtımlar ile ilgili kapsamlı bilgileri barındıran bir sitedir.

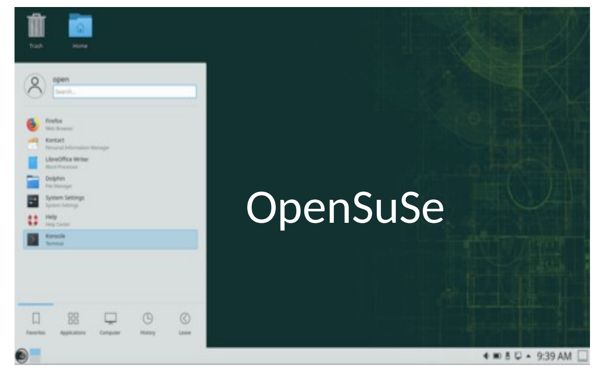
Ekim 2021

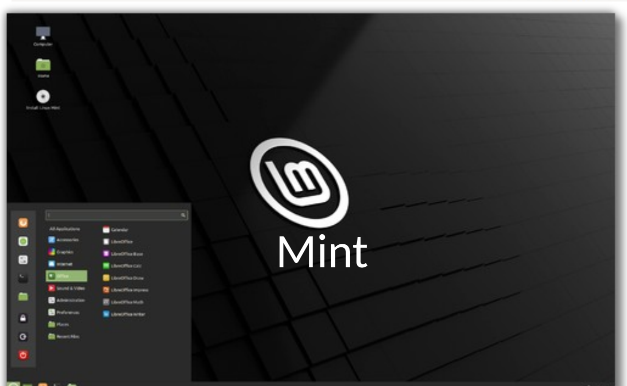
Last 12 months			Last 6 months			Last 3 months			Last 1 month		
1	MX Linux	3332▲	1	MX Linux	3177▲	1	MX Linux	3533▲	1	EndeavourOS	3741▼
2	Manjaro	2359▲	2	EndeavourOS	2745▲	2	EndeavourOS	3320▲	2	MX Linux	3666▲
3	EndeavourOS	2077▲	3	Manjaro	2234▲	3	Manjaro	2303–	3	Manjaro	2181▲
4	Mint	2054–	4	Mint	1933▲	4	Mint	1865▼	4	Mint	1955▲
5	Pop!_OS	1815▼	5	Pop!_OS	1540–	5	Ubuntu	1428▲	5	Ubuntu	1562▲
6	Ubuntu	1355–	6	Ubuntu	1317▲	6	Debian	1427▲	6	Pop!_OS	1494▲
7	Debian	1283▲	7	Debian	1277▲	7	Pop!_OS	1425▲	7	Debian	1308▲
8	elementary	1126▼	8	Garuda	1144▲	8	Garuda	1263–	8	Garuda	1271▼
9	Fedora	987–	9	elementary	1129–	9	elementary	1161–	9	Fedora	1155▲
10	Garuda	904–	10	Fedora	967–	10	Zorin	1016▲	10	Zorin	970▲
11	openSUSE	828–	11	Zorin	888▲	11	Fedora	933▲	11	elementary	956▼
12	Zorin	742▲	12	openSUSE	811▼	12	openSUSE	733–	12	openSUSE	712▼
13	KDE neon	738▼	13	KDE neon	699–	13	KDE neon	709▲	13	KDE neon	699▼
14	Solus	686▼	14	Solus	605–	14	Artix	629–	14	antiX	578▲
15	Arch	561▲	15	antiX	549▲	15	Lite	598▲	15	Kali	574▲
16	antiX	559▲	16	Arch	490▲	16	Solus	588▼	16	Lite	554▼
17	deepin	486▼	17	Slackware	461▼	17	antiX	543▲	17	Q4OS	552▲
18	Puppy	473–	18	Lite	449–	18	Slackware	539▲	18	Solus	541–
19	PCLinuxOS	460–	19	Artix	441–	19	PCLinuxOS	481▼	19	Slackware	538▲
20	Lite	445▲	20	PCLinuxOS	433–	20	Kali	432–	20	Feren	520▲
21	Kali	443–	21	Kali	414–	21	ArcoLinux	419▼	21	SparkyLinux	506▲
22	Slackware	425▲	22	Puppy	401–	22	SparkyLinux	417▲	22	Arch	445▲
23	ArcoLinux	424–	23	ArcoLinux	389–	23	Arch	400▲	23	Void	436▲
24	CentOS	420▼	24	deepin	364▼	24	Puppy	385–	24	Linuxfx	422▲
25	Mageia	410–	25	SparkyLinux	353▲	25	Q4OS	357▲	25	PCLinuxOS	420▲
26	FreeBSD	404▼	26	CentOS	350–	26	CentOS	319–	26	ExTiX	389▲
27	Linuxfx	400▼	27	Q4OS	348▲	27	Alpine	318–	27	PureOS	380▲
28	Q4OS	379–	28	FreeBSD	333▼	28	Tails	314▼	28	Puppy	377▲
29	SparkyLinux	348▲	29	Rocky	333▲	29	FreeBSD	313–	29	ArcoLinux	375▼
30	Artix	344–	30	Tails	313–	30	Devuan	310–	30	Artix	373▼
31	Alpine	342–	31	Mageia	312▼	31	Void	305▲	31	MidnightBSD	347▲
32	Ubuntu Kylin	339▼	32	Lubuntu	310–	32	Kaisen	301▼	32	Lubuntu	331▲
33	Lubuntu	328–	33	Alpine	305–	33	Lubuntu	299▲	33	Kubuntu	330▲
34	Tails	321–	34	Kubuntu	299–	34	Kubuntu	290▲	34	Archcraft	319▲
35	Kubuntu	304–	35	Devuan	296–	35	Bluestar	282▲	35	CentOS	317▲
36	Devuan	294–	36	RebornOS	290–	36	PureOS	276▲	36	Emmabuntüs	316–
37	Void	289▲	37	Bluestar	271▲	37	Mageia	272▼	37	Bluestar	311▲
38	Peppermint	288–	38	Linuxfx	270–	38	Feren	269▲	38	FreeBSD	301–
39	EasyOS	279▼	39	Void	269▲	39	Linuxfx	264–	39	Rocky	287▲
40	Bluestar	277▲	40	AlmaLinux	268–	40	Kodachi	262–	40	GhostBSD	284▼

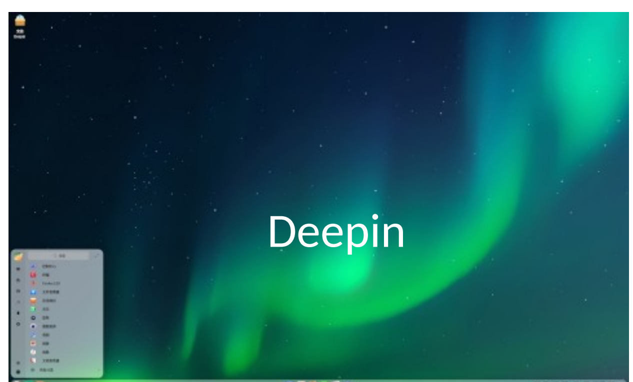
Çeşitli Dağıtımların Ekran Görüntüleri

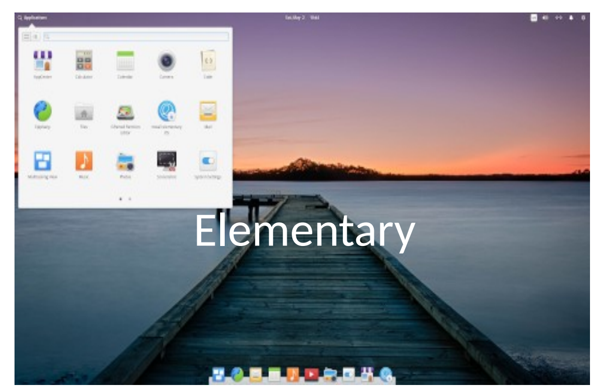












Siber Güvenlik Odaklı Linux Dağıtımları

- **Kali** Linux – www.kali.org
- **BlackArch** - www.blackarch.org
- **Parrot** Security – www.parrotlinux.org
- Komutan Linux – www.komutan.org
- Pentoo – www.pentoo.ch
- BackBox - www.backbox.org
- Demon Linux – www.demonlinux.com
- CAINE - www.caine-live.net
- ArchStrike - www.archstrike.org
- NST - www.networksecuritytoolkit.org

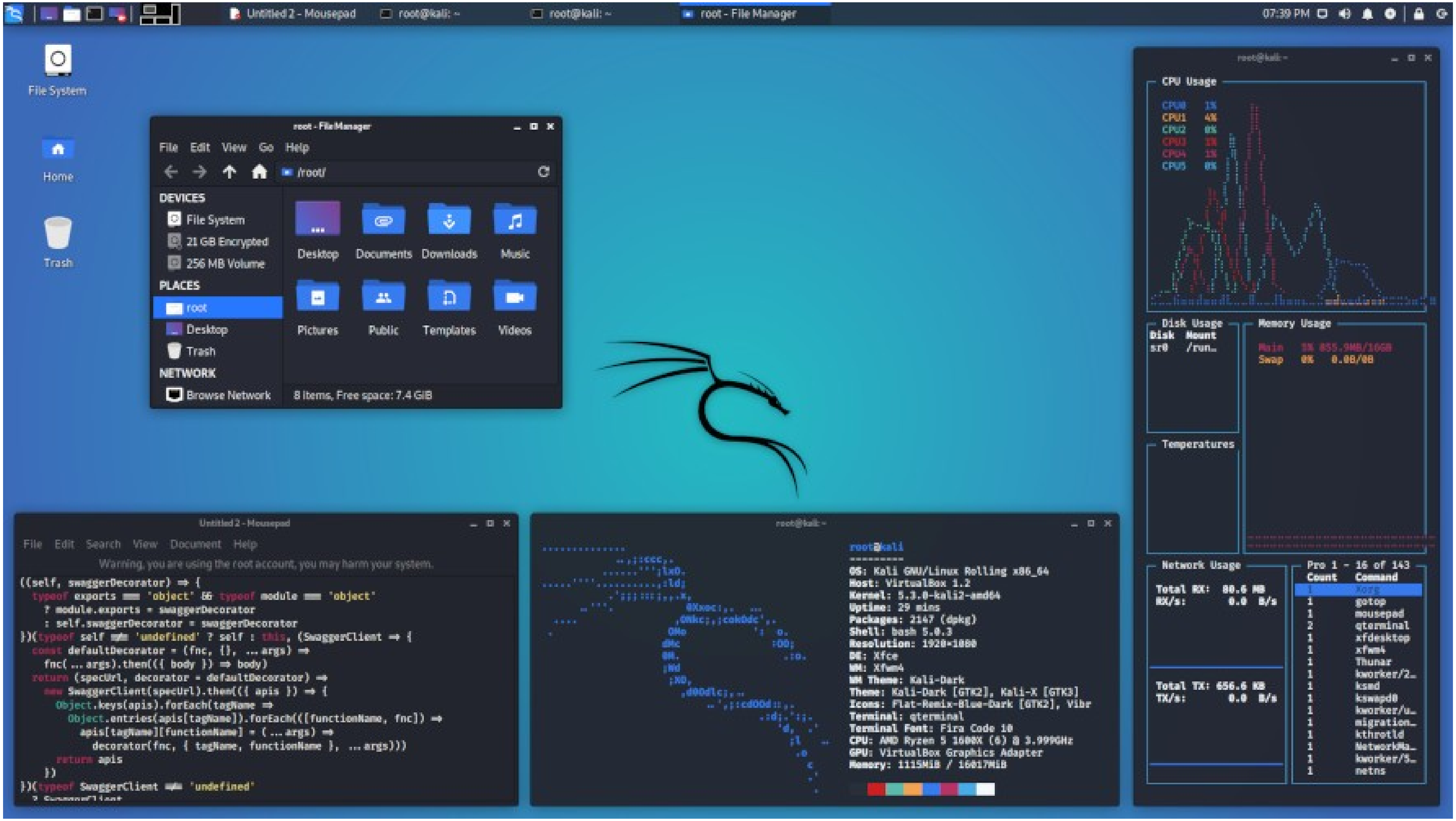
Kali Linux

- Penetrasyon Test ve Güvenlik Denetimi amaçlı bir Debian tabanlı Linux dağıtımıdır.
- Offensive Security Ltd. firması tarafından geliştirilmektedir.
- 2004 yılında BackTrack ile başlayan geliştirme,
- 2013 den itibaren Debian tabanı ile yeniden geliştirilmiştir ve Kali adı verilmiştir.

Linux Güvenlik Dağıtımları

Adı	**Kali Linux**
Sürümü	2021.3
Tabanı	Debian (BackTrack), [Abd, İsviçre]
Paket Yönetimi	Deb
Dağıtım Yöntemi	Rolling
Masaüstü	XFCE, GNOME, KDE Plasma, LXDE, MATE
Geliştirici	Offensive Security
İlk Sürüm Tarihi	2004
Web Sitesi	www.kali.org

Kali Linux (XFCE)

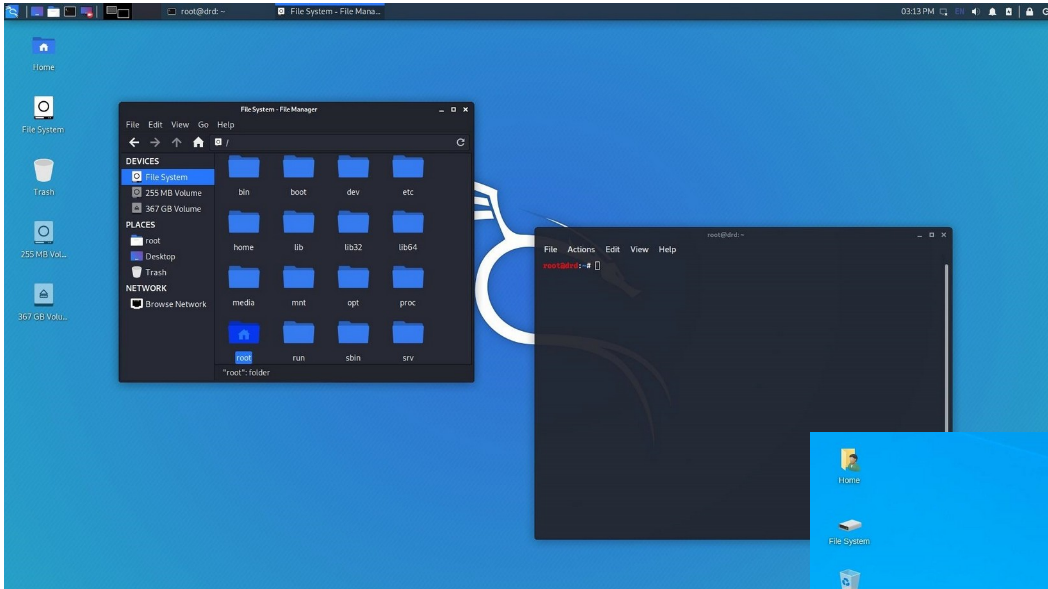


Kali Linux (GNOME)

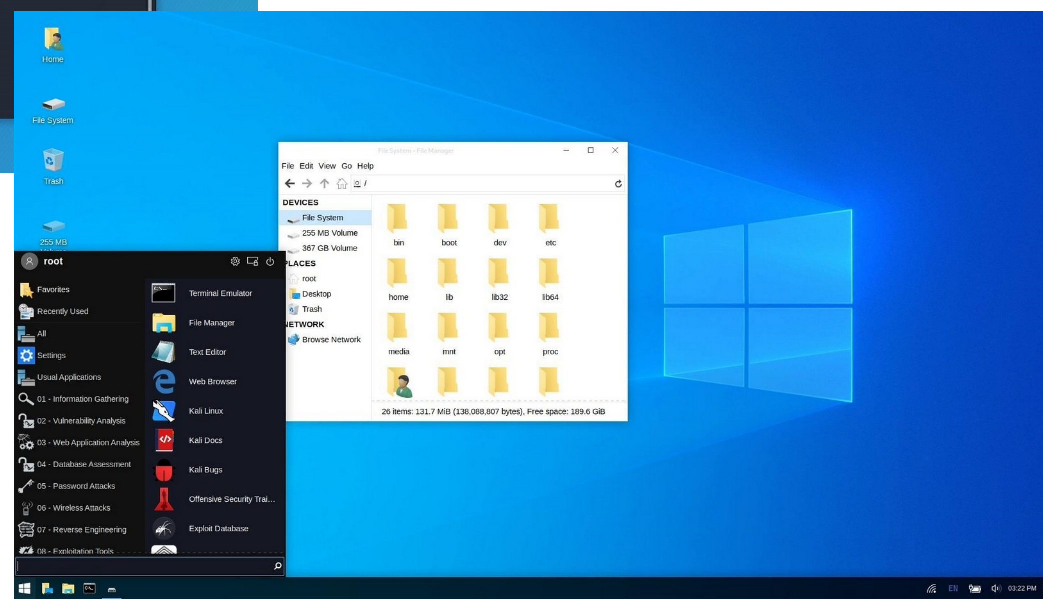
Kali Linux (KDE Plasma)

Kali Linux Undercover Mode:
Kali <--> Windows

Kali

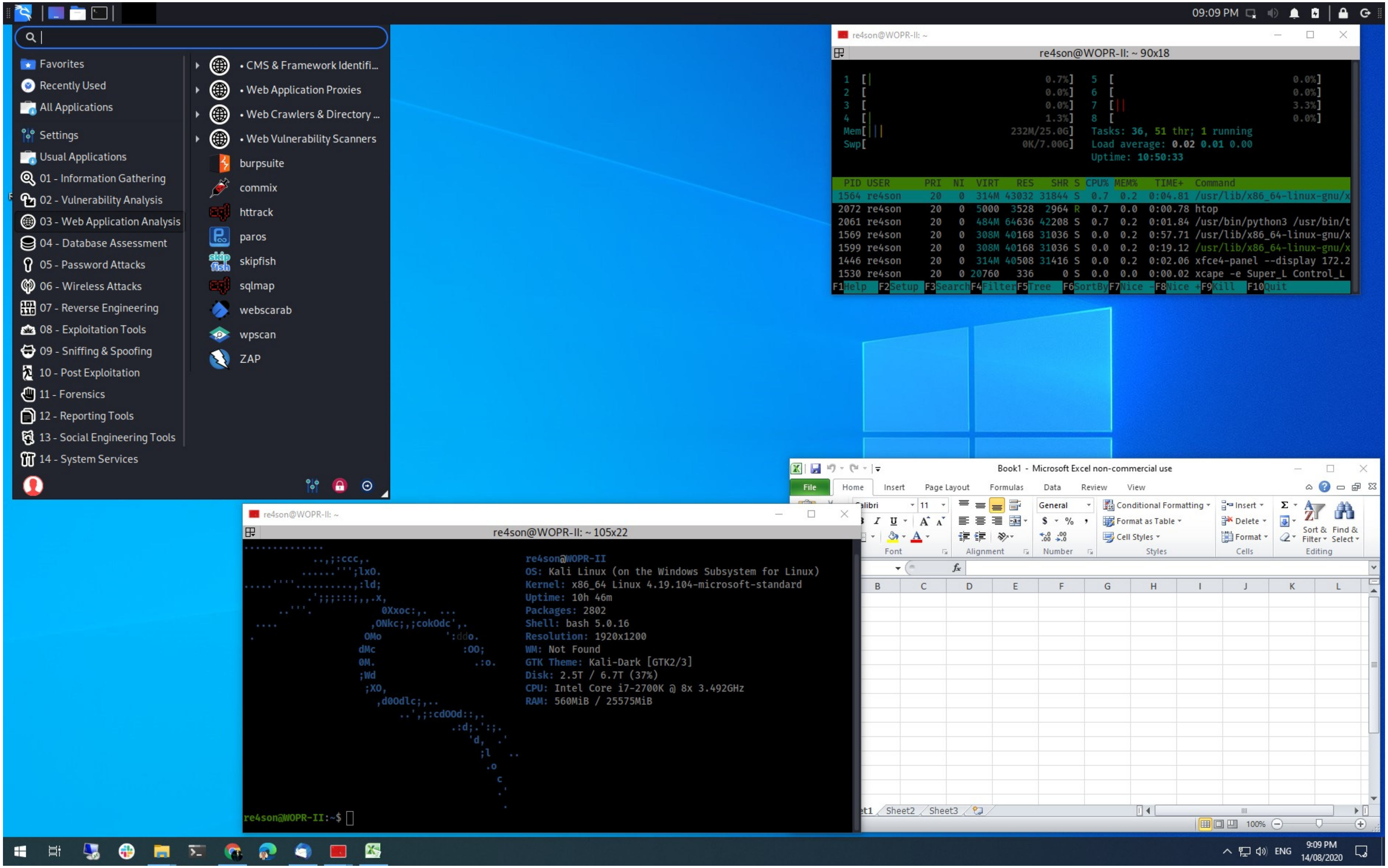


Windows

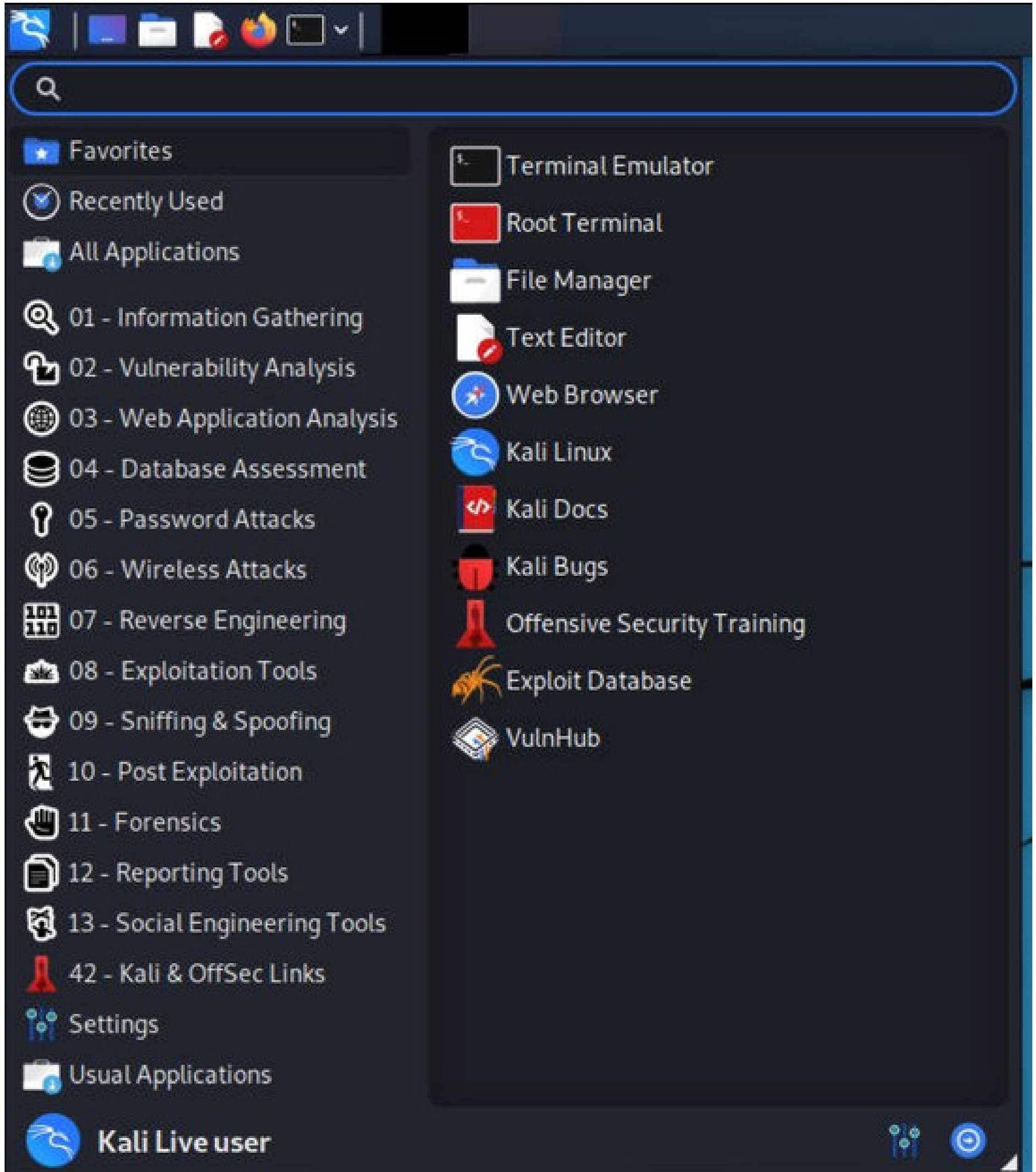


https://www.kali.org/docs/introduction/kali-undercover/

Kali Linux Win-KeX WSL2

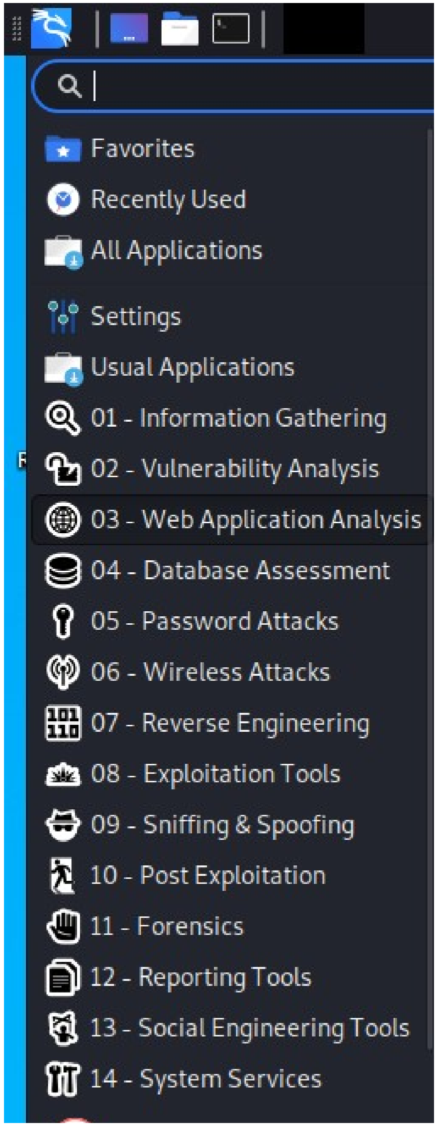


Kali Linux – Applications Menüsü



Kali Linux – Uygulamalar (Applications) Menüsü

1. Information Gathering (Bilgi Toplama Araçları)
2. Vulnerability Analysis (Zafiyet Tarama Araçları)
3. Web Applications Analysis (Web Güvenlik Açığı Tarayıcıları)
4. Database Assessment (Veritabanı Değerlendirmesi)
5. Password Atatacks (Şifre Atakları)
6. Wireless Attacks (Kablosuz Ağ Atakları)
7. Reverse Engineering (Tersine Mühendislik)
8. Exploitation Tools (Sömürü Araçları)
9. Sniffing & Spoofing (Koklama ve Sahtecilik)
10. Post Exploitation (Sömürü Sonrası)
11. Forensics (Adli Bilişim Araçları)
12. Reporting Tools (Raporlama Araçları)
13. Social Engineering Tools (Sosyal Mühendislik Araçları)
14. System Service (Sistem Hizmeti)



Kali Linux - Araçlar

Information Gathering

- acccheck
- ace-voip
- Amap
- Automater
- bing-ip2hosts
- braa
- CaseFile
- CDPSnarf
- cisco-torch
- Cookie Cadger
- copy-router-config
- DMitry
- dnmap
- dnsenum
- dnsmap
- DNSRecon
- dnstracer
- dnswalk
- DotDotPwn
- enum4linux
- enumIAX
- Faraday
- Fierce
- Firewalk
- fragroute

Vulnerability Analysis

- BBQSQL
- BED
- cisco-auditing-tool
- cisco-global-exploiter
- cisco-ocs
- cisco-torch
- copy-router-config
- DBPwAudit
- Doona
- DotDotPwn
- HexorBase
- Inguma
- jSQL
- Lynis
- Nmap
- ohrwurm
- Oscanner
- Powerfuzzer
- sfuzz
- SidGuesser
- SIPArmyKnife
- sqlmap
- Sqlninja
- sqlsus
- THC-IPV6

Wireless Attacks

- Aircrack-ng
- Asleap
- Bluelog
- BlueMaho
- Bluepot
- BlueRanger
- Bluesnarfer
- Bully
- coWPAtty
- crackle
- eapmd5pass
- Fern Wifi Cracker
- Ghost Phisher
- GISKismet
- Gqrx
- gr-scan
- hostapd-wpe
- kalibrate-rtl
- KillerBee
- Kismet
- mdk3
- mfcuk
- mfoc
- mfterm
- Multimon-NG

Web Applications

- apache-users
- Arachni
- BBQSQL
- BlindElephant
- Burp Suite
- CutyCapt
- DAVTest
- deblaze
- DIRB
- DirBuster
- fimap
- FunkLoad
- Gobuster
- Grabber
- jboss-autopwn
- joomscan
- jSQL
- Maltego Teeth
- PadBuster
- Paros
- Parsero
- plecost
- Powerfuzzer
- ProxyStrike
- Recon-ng

https://tools.kali.org/tools-listing

Kali Linux - Araçlar

- fragrouter
- Ghost Phisher
- GoLismero
- goofile
- hping3
- ident-user-enum
- InTrace
- iSMTP
- lbd
- Maltego Teeth
- masscan
- Metagoofil
- Miranda
- nbtscan-unixwiz
- Nmap
- ntop
- p0f
- Parsero
- Recon-ng
- SET
- smtp-user-enum
- snmp-check
- SPARTA
- sslcaudit
- SSLsplit
- sslstrip
- SSLyze
- THC-IPV6
- theHarvester
- tnscmd10g
- unix-privesc-check
- Yersinia

Exploitation Tools

- Armitage
- Backdoor Factory
- BeEF
- cisco-auditing-tool
- cisco-global-exploiter
- cisco-ocs
- cisco-torch
- Commix
- crackle
- exploitdb
- jboss-autopwn
- Linux Exploit Suggester
- Maltego Teeth
- Metasploit Framework
- RouterSploit
- SET
- ShellNoob
- sqlmap
- THC-IPV6
- Yersinia

- PixieWPS
- Reaver
- redfang
- RTLSDR Scanner
- Spooftooph
- Wifi Honey
- wifiphisher
- Wifitap
- Wifite

Forensics Tools

- Binwalk
- bulk-extractor
- Capstone
- chntpw
- Cuckoo
- dc3dd
- ddrescue
- DFF
- diStorm3
- Dumpzilla
- extundelete
- Foremost
- Galleta
- Guymager
- iPhone Backup Analyzer
- p0f

- Skipfish
- sqlmap
- Sqlninja
- sqlsus
- ua-tester
- Uniscan
- Vega
- w3af
- WebScarab
- Webshag
- WebSlayer
- WebSploit
- Wfuzz
- WPScan
- XSSer
- zaproxy

Stress Testing

- DHCPig
- FunkLoad
- iaxflood
- Inundator
- inviteflood
- ipv6-toolkit
- mdk3
- Reaver
- rtpflood

https://tools.kali.org/tools-listing

Kali Linux - Araçlar

Sniffing & Spoofing

- Burp Suite
- DNSChef
- fiked
- hamster-sidejack
- HexInject
- iaxflood
- inviteflood
- iSMTP
- isr-evilgrade
- mitmproxy
- ohrwurm
- protos-sip
- rebind
- responder
- rtpbreak
- rtpinsertsound
- rtpmixsound
- sctpscan
- SIPArmyKnife
- SIPp
- SIPVicious
- SniffJoke
- SSLsplit
- sslstrip
- THC-IPV6
- VoIPHopper

Password Attacks

- acccheck
- Burp Suite
- CeWL
- chntpw
- cisco-auditing-tool
- CmosPwd
- creddump
- crunch
- DBPwAudit
- findmyhash
- gpp-decrypt
- hash-identifier
- HexorBase
- THC-Hydra
- John the Ripper
- Johnny
- keimpx
- Maltego Teeth
- Maskprocessor
- multiforcer
- Ncrack
- oclgausscrack
- PACK
- patator
- phrasendrescher
- polenum

Maintaining Access

- CryptCat
- Cymothoa
- dbd
- dns2tcp
- http-tunnel
- HTTPTunnel
- Intersect
- Nishang
- polenum
- PowerSploit
- pwnat
- RidEnum
- sbd
- U3-Pwn
- Webshells
- Weevely
- Winexe

Hardware Hacking

- android-sdk
- apktool
- Arduino
- dex2jar
- Sakis3G

Reverse Engineering

- apktool
- dex2jar
- diStorm3
- edb-debugger
- jad
- javasnoop
- JD-GUI
- OllyDbg
- smali
- Valgrind
- YARA

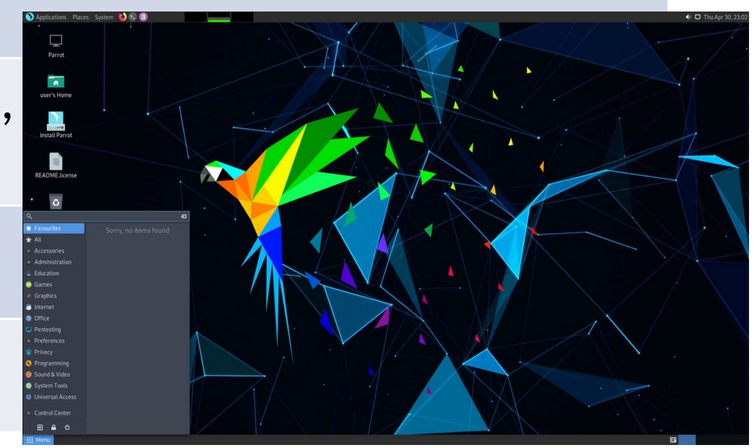
Reporting Tools

- CaseFile
- CutyCapt
- dos2unix
- Dradis
- KeepNote
- MagicTree
- Metagoofil
- Nipper-ng
- pipal

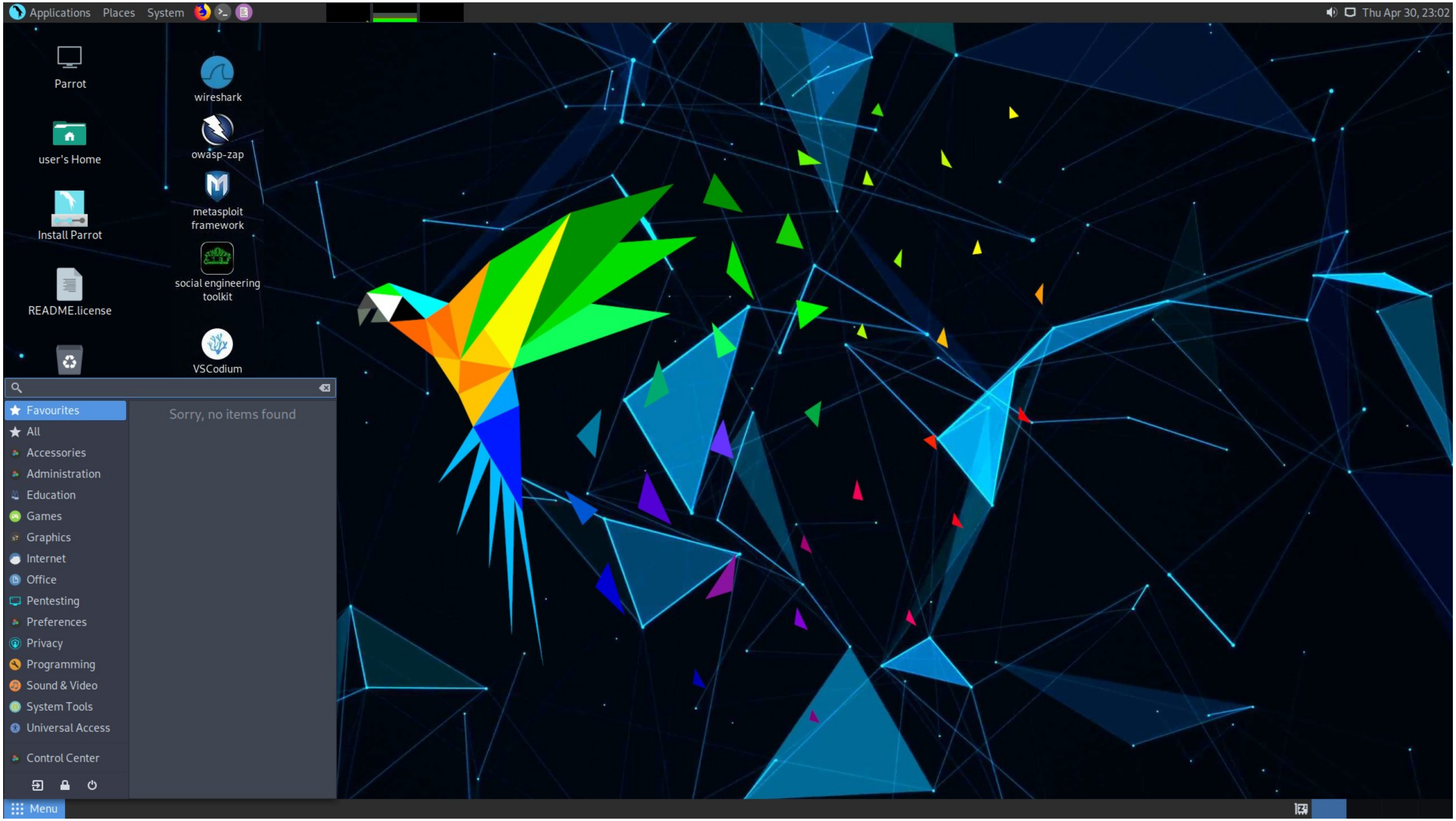
https://tools.kali.org/tools-listing

Linux Güvenlik Dağıtımları

Adı	**Parrot OS** (Parrot Security)
Sürümü	4.11
Tabanı	Debian, [İtalya]
Paket Yönetimi	Deb (apt)
Dağıtım Yöntemi	Fixed
Masaüstü	KDE Plasma, MATE
Geliştirici	Lorenzo Faletra (palinuro), Parrot Dev Team
İlk Sürüm Tarihi	2013
Web Sitesi	www.parrotsec.org



Parrot OS (Parrot Security)

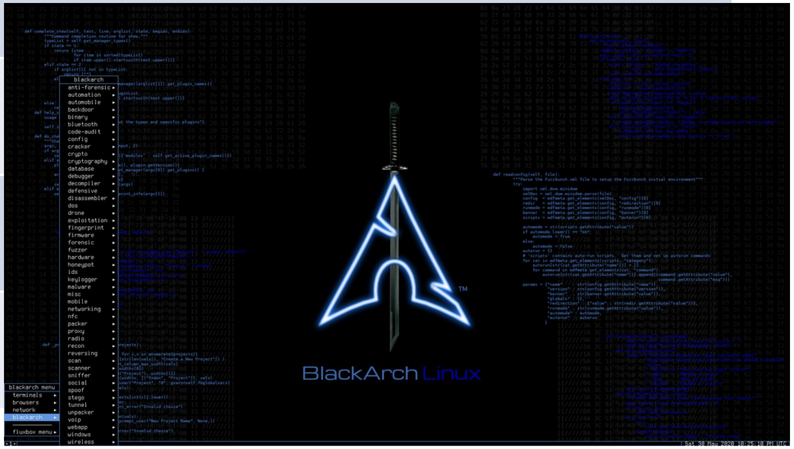


www.parrotlinux.org

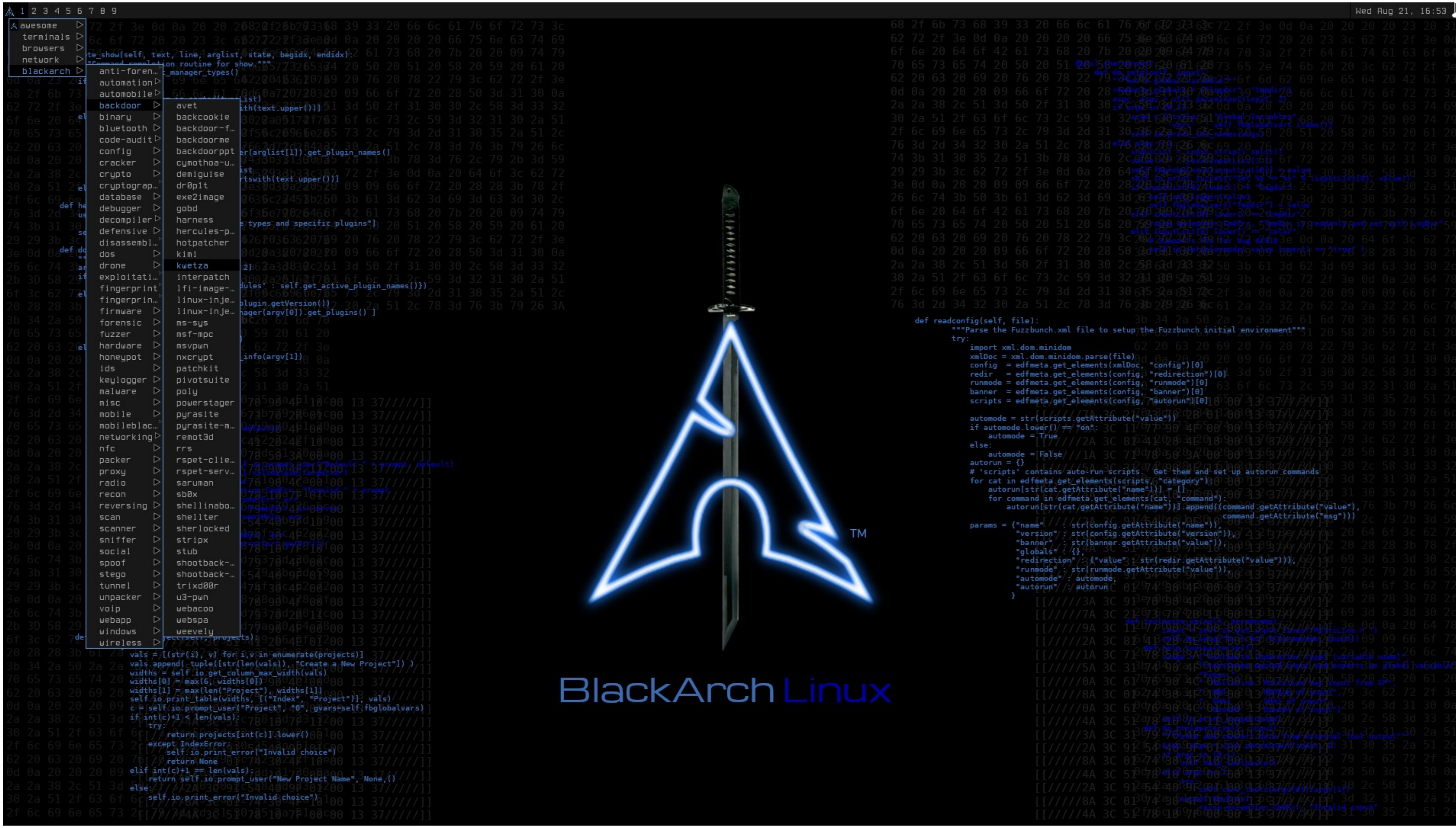
Linux Güvenlik Dağıtımları

BlackArch Linux

Adı	BlackArch Linux
Sürümü	2021.09.1
Tabanı	Arch Linux [Amerika]
Paket Yönetimi	Pacman
Dağıtım Yöntemi	Fixed
Masaüstü	Fluxbox
Geliştirici	Gönüllü Ekibi
İlk Sürüm Tarihi	2013
Web Sitesi	www.blackarch.org



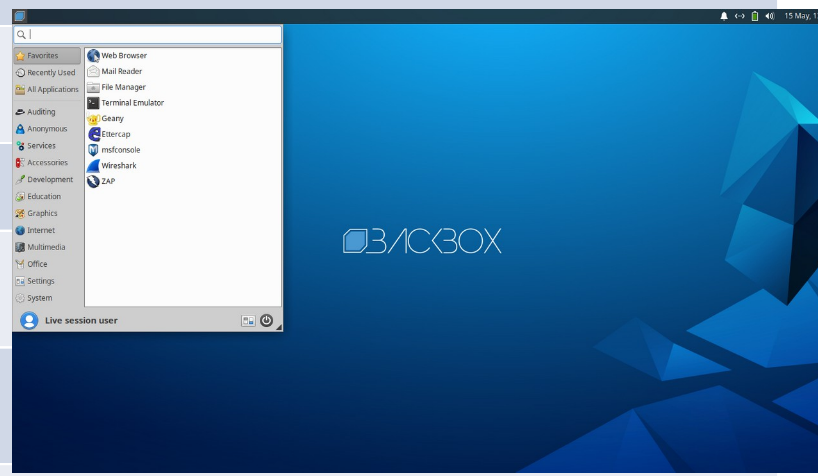
BlackArch Linux



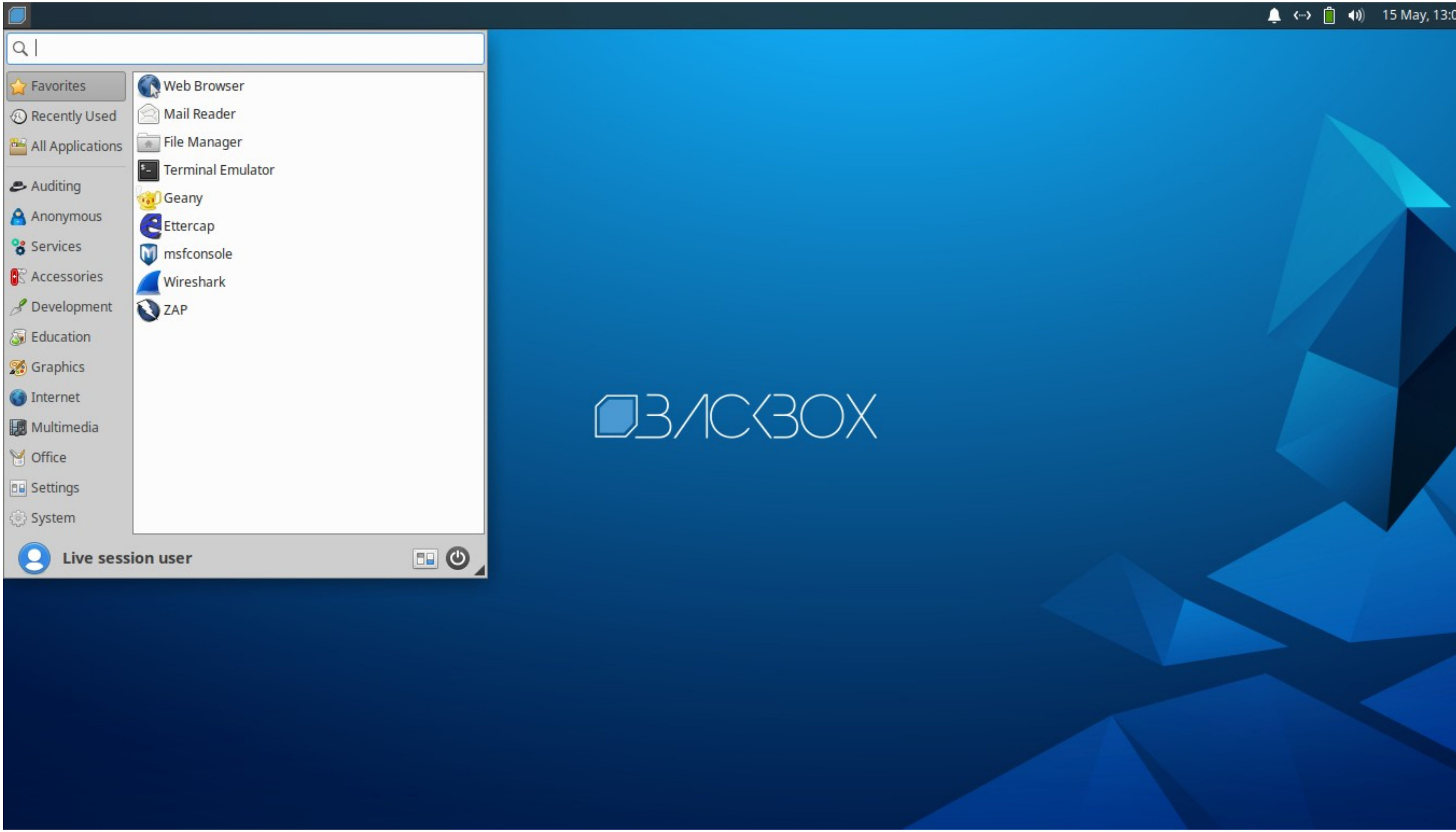
www.blackarch.org

Linux Güvenlik Dağıtımları

Adı	BackBox Linux
Sürümü	7 (2020-05-15)
Tabanı	Debian, Ubuntu [İtalya]
Paket Yönetimi	Deb
Dağıtım Yöntemi	Fixed
Masaüstü	XFCE
Geliştirici	BackBox Team
İlk Sürüm Tarihi	2004
Web Sitesi	https://linux.backbox.org



BackBox Linux

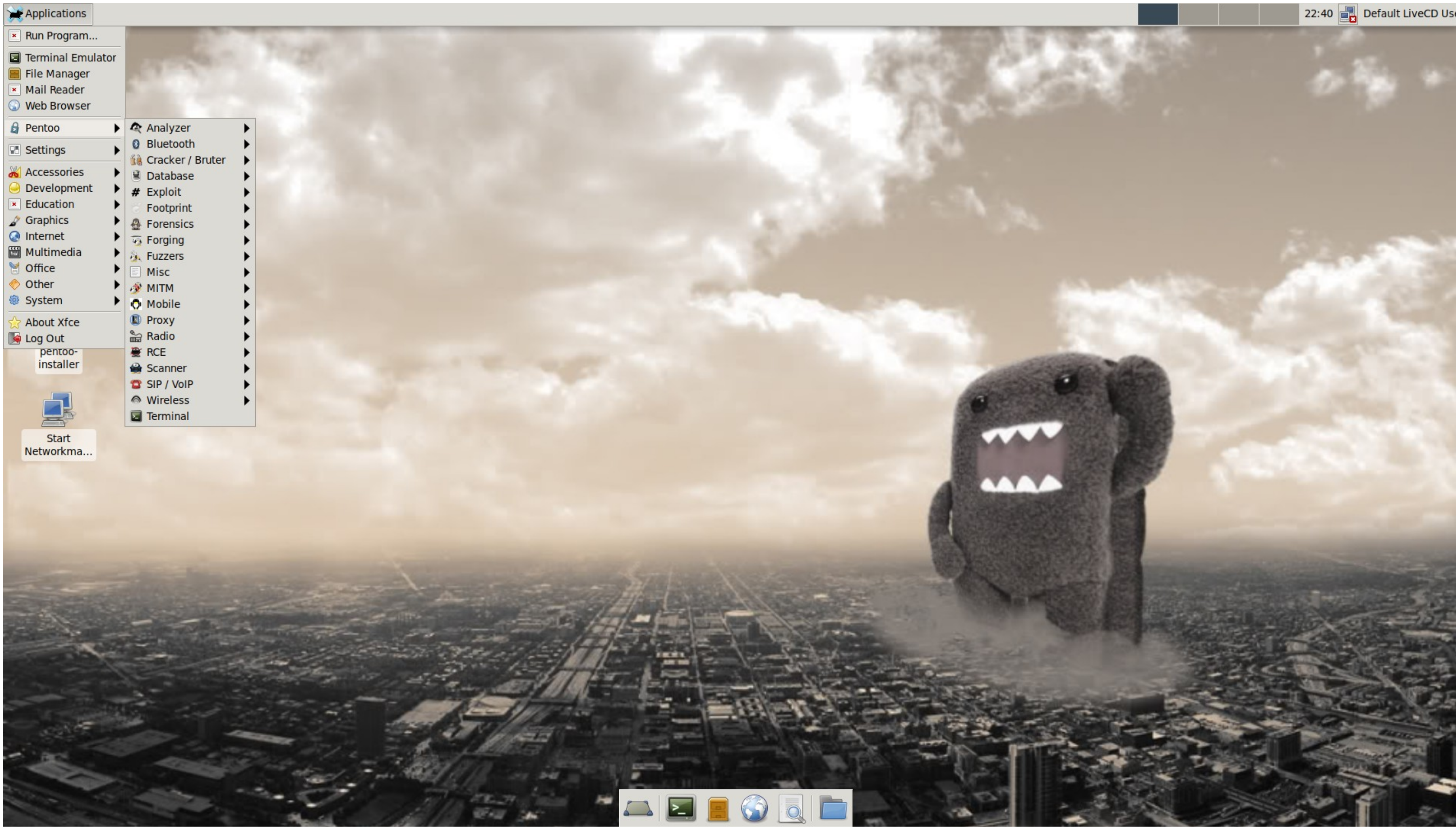


https://linux.backbox.org

Linux Güvenlik Dağıtımları

Adı	Pentoo Linux
Sürümü	2020.0 (2021.0 günlük)
Tabanı	Gentoo [İsviçre]
Paket Yönetimi	Portage
Dağıtım Yöntemi	Rolling
Masaüstü	XFCE
Geliştirici	Pentoo Ekibi
İlk Sürüm Tarihi	2005
Web Sitesi	www.pentoo.ch

Pentoo Linux



https://linux.backbox.org

Linux Güvenlik Dağıtımları

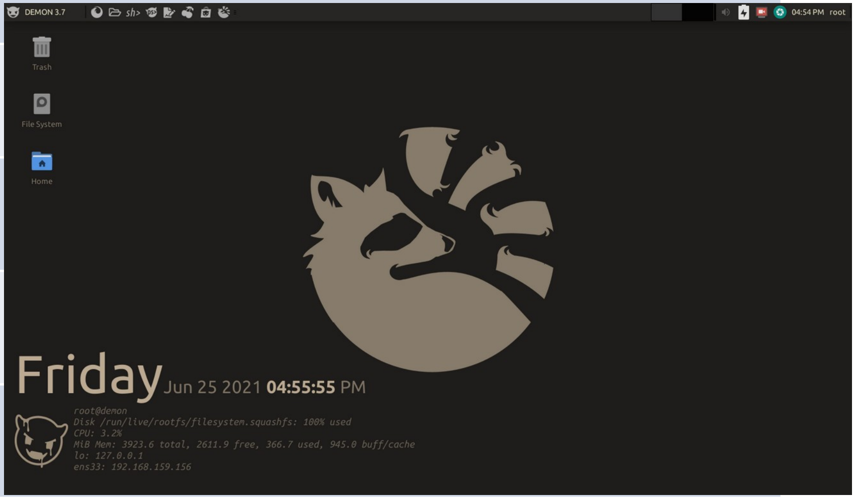
Adı	Komutan Linux
Sürümü	1.0 Beta
Tabanı	~~Debian~~, Milis Linux [Türkiye]
Paket Yönetimi	Apt
Dağıtım Yöntemi	?
Masaüstü	XFCE4
Geliştirici	Aydın Yakar, Kubilay Güngör
İlk Sürüm Tarihi	2018 Kasım, (Rc 2017)
Web Sitesi	www.komutan.org

Komutan Linux

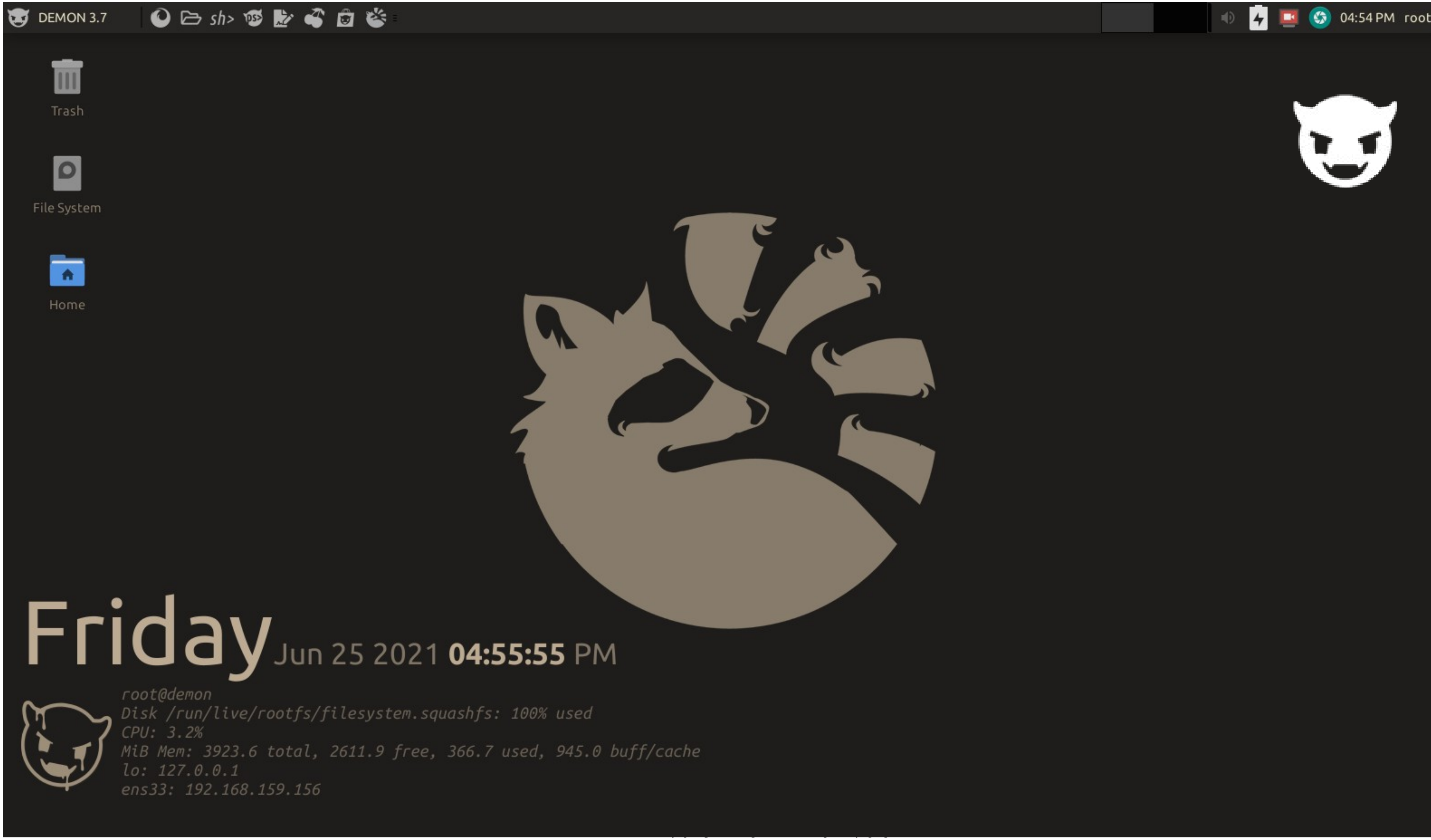
www.komutan.org

Linux Güvenlik Dağıtımları

Adı	Demon Linux
Sürümü	3.7.1
Tabanı	Debian (WeakNet Linux)
Paket Yönetimi	Apt
Dağıtım Yöntemi	?
Masaüstü	XFCE4
Geliştirici	WeakNet Labs
İlk Sürüm Tarihi	?
Web Sitesi	www.demonlinux.com

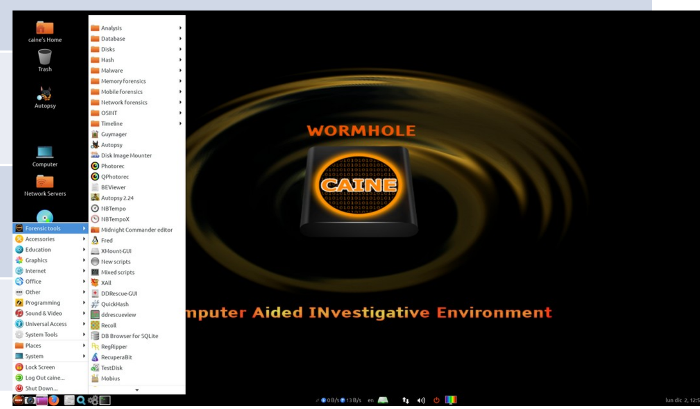


Demon Linux

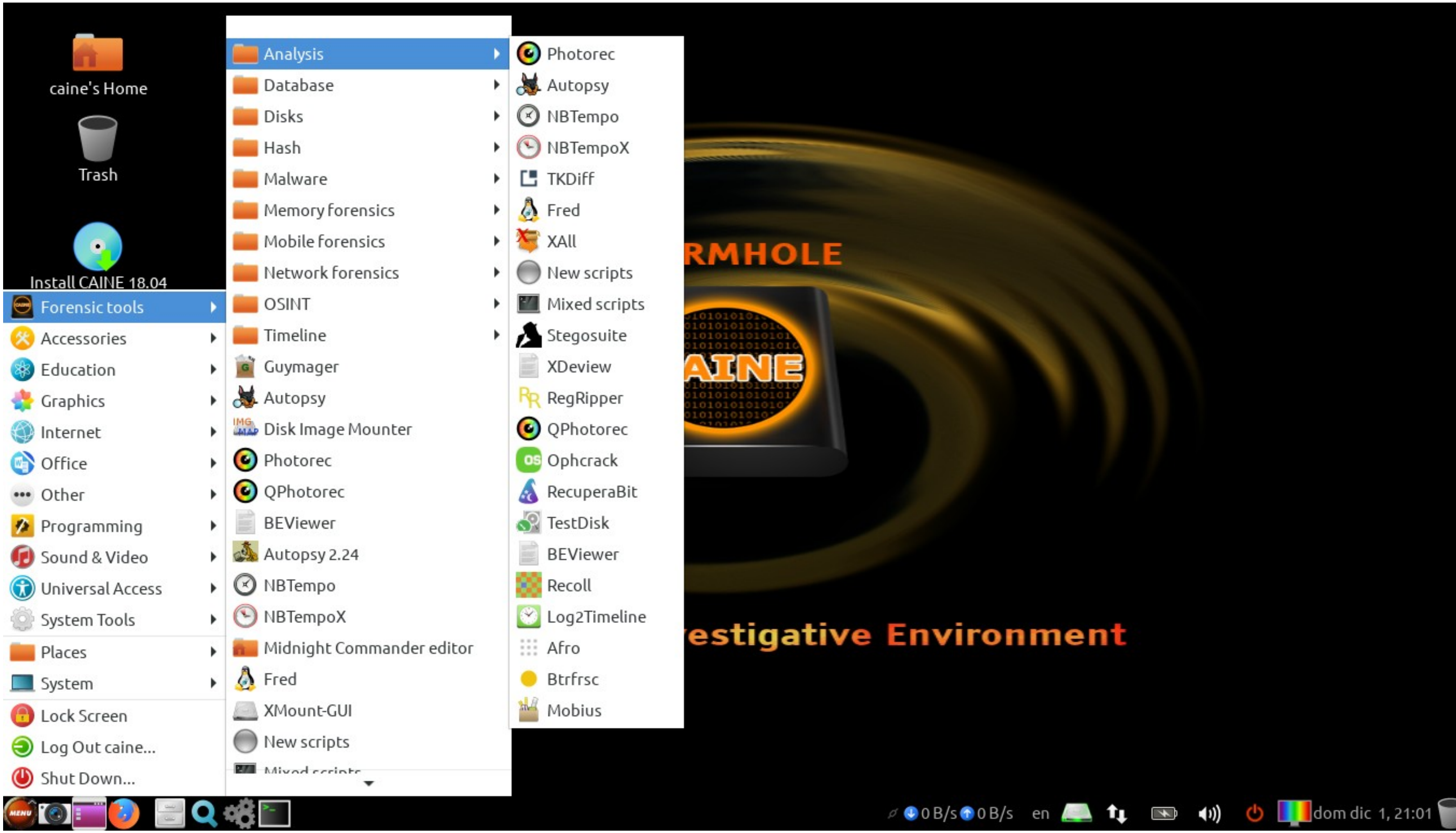

DEMON 3.7
04:54 PM root
Trash
File System
Home
Friday Jun 25 2021 04:55:55 PM
root@demon
Disk /run/live/rootfs/filesystem.squashfs: 100% used
CPU: 3.2%
MiB Mem: 3923.6 total, 2611.9 free, 366.7 used, 945.0 buff/cache
lo: 127.0.0.1
ens33: 192.168.159.156

Linux Güvenlik Dağıtımları

Adı	**CAINE** (**C**omputer **A**ided **IN**vestigative **E**nvironment)
Sürümü	11.0 (Wormhole)
Tabanı	Debian, Ubuntu [İtalya]
Paket Yönetimi	Deb
Dağıtım Yöntemi	Fixed
Masaüstü	Mate
Geliştirici	Nanni Bassetti
İlk Sürüm Tarihi	2009
Web Sitesi	www.caine-live.net



CAINE



www.caine-live.net

Linux Güvenlik Dağıtımları

ArchStrike

Adı	**ArchStrike**
Sürümü	2020.09.18
Tabanı	Arch Linux [Amerika]
Paket Yönetimi	Pacman
Dağıtım Yöntemi	Rolling
Masaüstü	Openbox
Geliştirici	Vincent Loup, Oğuz Bektaş, Michael Henze, Kevin MacMartin, James Stronz, Chad Seaman
İlk Sürüm Tarihi	?
Web Sitesi	www.archstrike.org

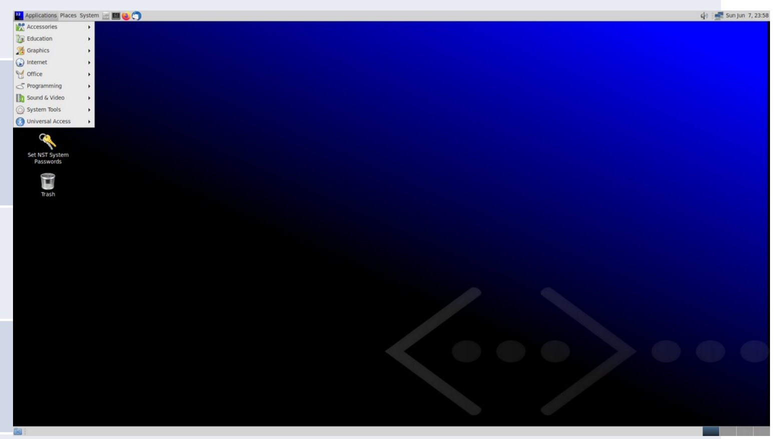
ArchStrike

ArchStrike

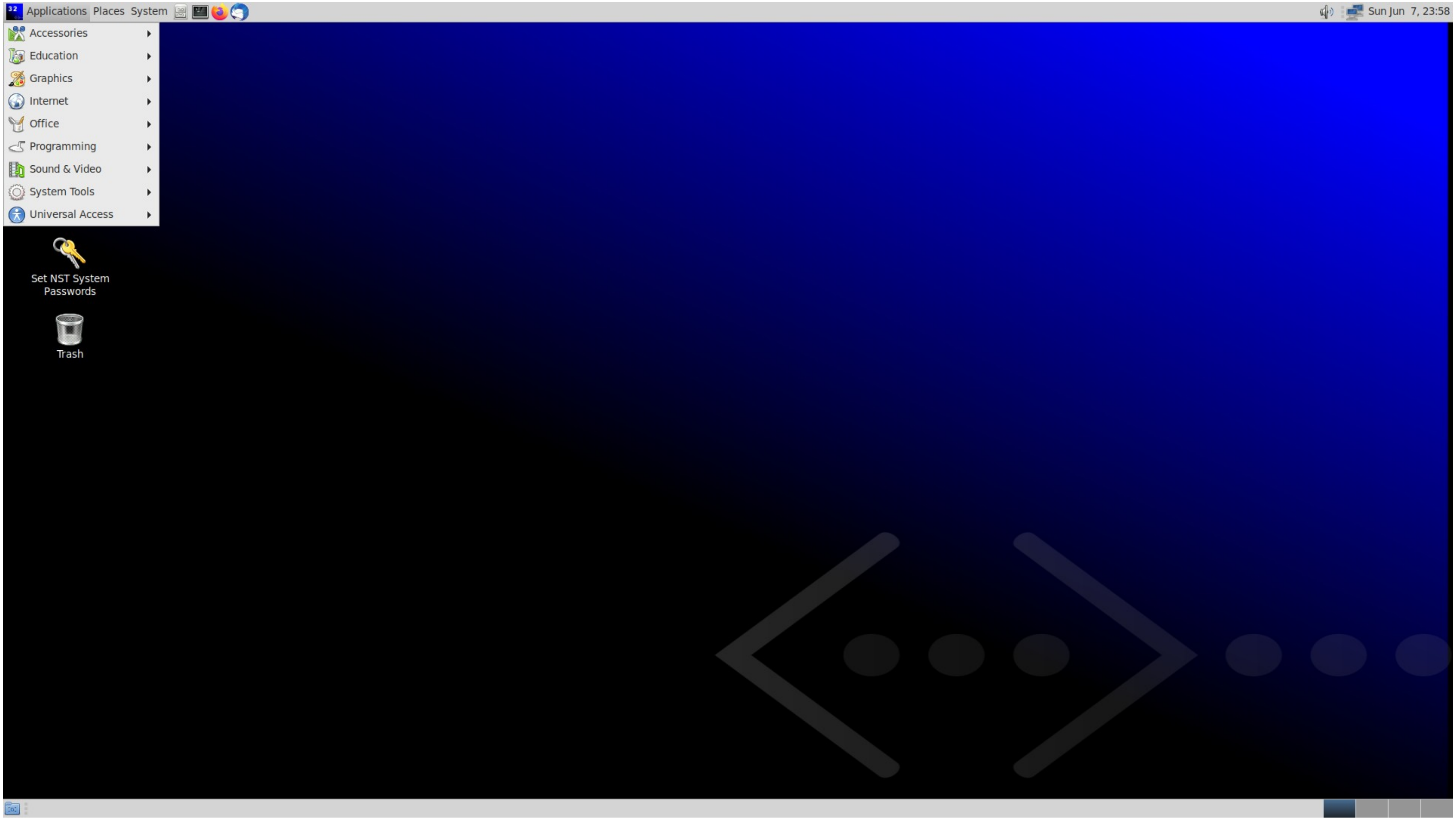
www.archstrike.org

Linux Güvenlik Dağıtımları

Adı	NST (Network Security Toolkit)
Sürümü	34-12783 [Live] (2021-8)
Tabanı	Fedora [Amerika]
Paket Yönetimi	Rpm
Dağıtım Yöntemi	Fixed
Masaüstü	Mate, Fluxbox, Openbox
Geliştirici	NST ?
İlk Sürüm Tarihi	2005
Web Sitesi	www.networksecuritytoolkit.org



NST (Network Security Toolkit)



www.networksecuritytoolkit.org

Timeline of Windows

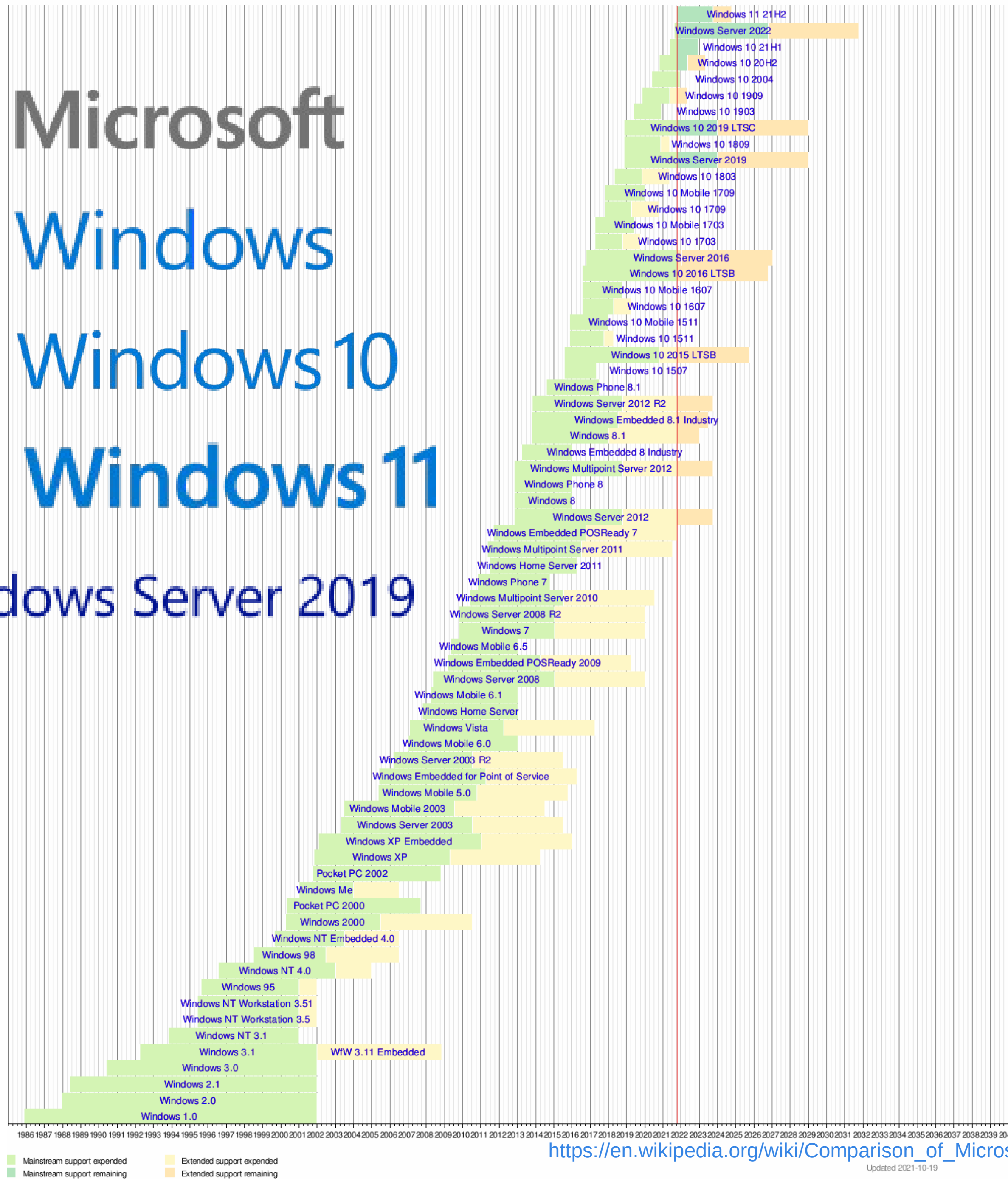
Microsoft

Windows

Windows 10

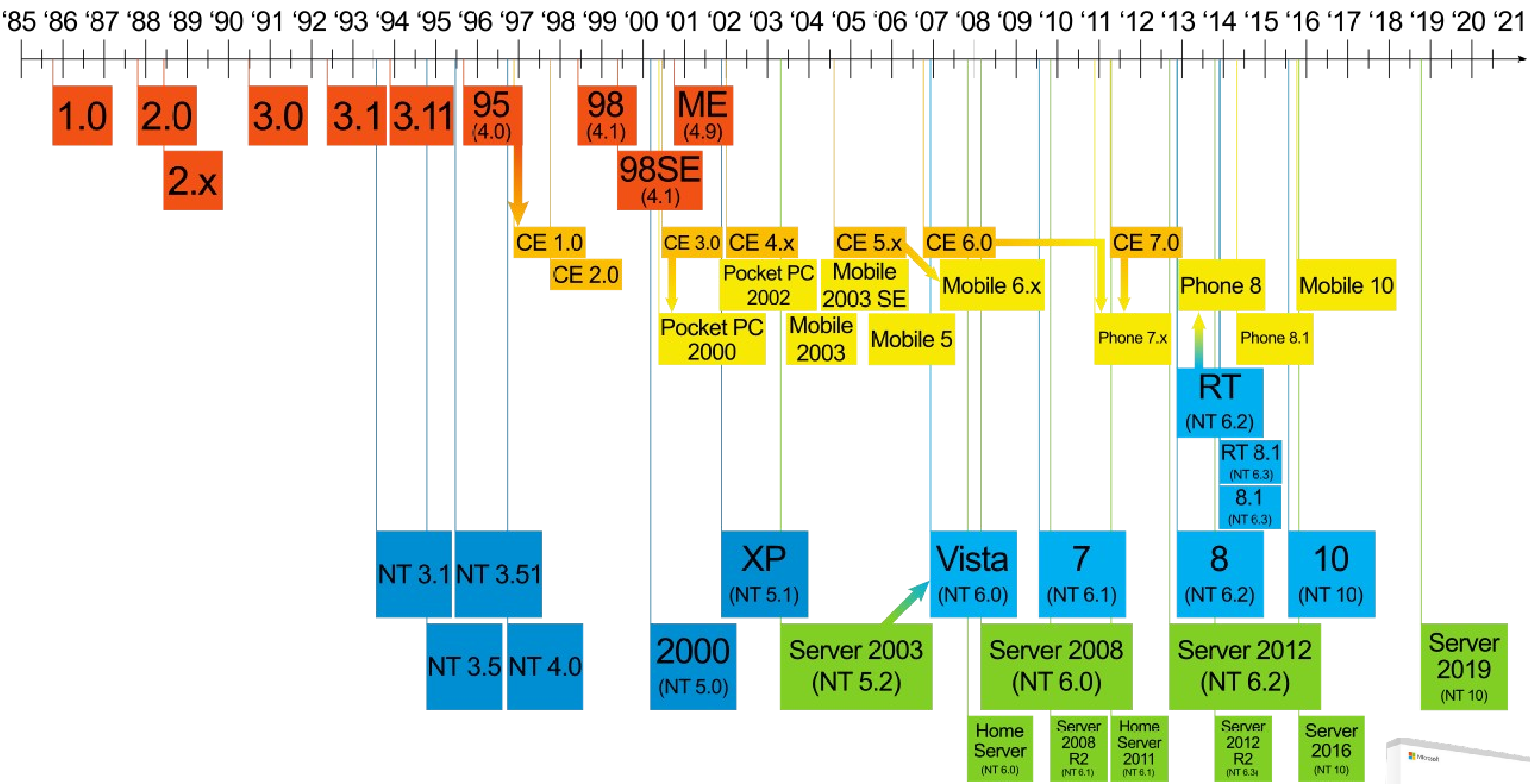
Windows 11

Windows Server 2019



https://en.wikipedia.org/wiki/Comparison_of_Microsoft_Windows_versions

Microsoft Windows



Windows 11 Windows Server 2019

Mac OS X, OS X, and macOS version information

<table>
<tr><th>Version</th><th>Codename</th><th>Darwin version</th><th>Processor support</th><th>Application support</th><th>Kernel</th><th>Date announced</th><th>Release date</th><th>End of support date</th><th>Most recent version</th></tr>
<tr><td>Rhapsody Developer Release</td><td>Grail1Z4 / Titan1U</td><td></td><td rowspan="6">32-bit PowerPC</td><td rowspan="8">32-bit PowerPC</td><td rowspan="10">32-bit</td><td>January 7, 1997[159]</td><td>August 31, 1997</td><td>Unknown</td><td>DR2
(May 14, 1998)</td></tr>
<tr><td>Mac OS X Server 1.0</td><td>Hera</td><td></td><td>Unknown</td><td>March 16, 1999</td><td>Unknown</td><td>1.2v3
(October 27, 2000)</td></tr>
<tr><td>Mac OS X Developer Preview</td><td>Unknown</td><td></td><td>May 11, 1998[160]</td><td>March 16, 1999</td><td>Unknown</td><td>DP4
(April 5, 2000)</td></tr>
<tr><td>Mac OS X Public Beta</td><td>Kodiak[161]</td><td></td><td>May 15, 2000[162]</td><td>September 13, 2000</td><td>March 24, 2001[citation needed]</td><td>N/A</td></tr>
<tr><td>Mac OS X 10.0</td><td>Cheetah</td><td>1.3.1</td><td>January 9, 2001[163]</td><td>March 24, 2001</td><td>2004[citation needed]</td><td>10.0.4 (4Q12)
(June 22, 2001)</td></tr>
<tr><td>Mac OS X 10.1</td><td>Puma</td><td>1.4.1 / 5</td><td>July 18, 2001[164]</td><td>September 25, 2001</td><td>2005[citation needed]</td><td>10.1.5 (5S60)
(June 6, 2002)</td></tr>
<tr><td>Mac OS X 10.2</td><td>Jaguar</td><td>6</td><td>32/64-bit PowerPC[Note 1]</td><td>May 6, 2002[165]</td><td>August 24, 2002</td><td>2006[citation needed]</td><td>10.2.8
(October 3, 2003)</td></tr>
<tr><td>Mac OS X 10.3</td><td>Panther</td><td>7</td><td>32/64-bit PowerPC</td><td>June 23, 2003[166]</td><td>October 24, 2003</td><td>2007[citation needed]</td><td>10.3.9 (7W98)
(April 15, 2005)</td></tr>
<tr><td>Mac OS X 10.4</td><td>Tiger</td><td>8</td><td rowspan="2">32/64-bit PowerPC and Intel</td><td>32/64-bit[Note 2] PowerPC[Note 3] and Intel</td><td>May 4, 2004[167]</td><td>April 29, 2005</td><td>September 2009[citation needed]</td><td>10.4.11
(November 14, 2007)</td></tr>
<tr><td>Mac OS X 10.5</td><td>Leopard</td><td>9</td><td>32/64-bit PowerPC[Note 3] and Intel</td><td>June 26, 2006[168]</td><td>October 26, 2007</td><td>June 23, 2011[citation needed]</td><td>10.5.8 (9L31a)
(August 13, 2009)</td></tr>
<tr><td>Mac OS X 10.6</td><td>Snow Leopard</td><td>10</td><td>32/64-bit Intel</td><td>32/64-bit Intel
32-bit PowerPC[Note 3]</td><td rowspan="2">32/64-bit[169]</td><td>June 9, 2008[170]</td><td>August 28, 2009</td><td>February 25, 2014[citation needed]</td><td>10.6.8 (10K549)
(July 25, 2011)</td></tr>
<tr><td>Mac OS X 10.7</td><td>Lion</td><td>11</td><td rowspan="9">64-bit Intel</td><td rowspan="8">32/64-bit Intel</td><td>October 20, 2010[171]</td><td>July 20, 2011</td><td>October 2014[citation needed]</td><td>10.7.5 (11G63)
(October 4, 2012)</td></tr>
<tr><td>OS X 10.8</td><td>Mountain Lion</td><td>12</td><td rowspan="10">64-bit[172]</td><td>February 16, 2012[173]</td><td>July 25, 2012[174]</td><td>September 2015[citation needed]</td><td>10.8.5 (12F2560)
(August 13, 2015)</td></tr>
<tr><td>OS X 10.9</td><td>Mavericks</td><td>13</td><td>June 10, 2013[175]</td><td>October 22, 2013</td><td>September 2016[citation needed]</td><td>10.9.5 (13F1911)
(July 18, 2016)</td></tr>
<tr><td>OS X 10.10</td><td>Yosemite</td><td>14</td><td>June 2, 2014[176]</td><td>October 16, 2014</td><td>August 2017[citation needed]</td><td>10.10.5 (14F2511)
(July 19, 2017)</td></tr>
<tr><td>OS X 10.11</td><td>El Capitan</td><td>15</td><td>June 8, 2015[177]</td><td>September 30, 2015</td><td>September 2018[citation needed]</td><td>10.11.6 (15G22010)
(July 9, 2018)</td></tr>
<tr><td>macOS 10.12</td><td>Sierra</td><td>16</td><td>June 13, 2016[178]</td><td>September 20, 2016</td><td>October 2019[citation needed]</td><td>10.12.6 (16G2136)
(September 26, 2019)</td></tr>
<tr><td>macOS 10.13</td><td>High Sierra</td><td>17</td><td>June 5, 2017</td><td>September 25, 2017</td><td>December 2020[citation needed]</td><td>10.13.6 (17G14042)
(November 12, 2020)</td></tr>
<tr><td>macOS 10.14</td><td>Mojave</td><td>18</td><td>June 4, 2018</td><td>September 24, 2018</td><td>TBA</td><td>10.14.6 (18G9323)
(July 21, 2021)</td></tr>
<tr><td>macOS 10.15</td><td>Catalina</td><td>19</td><td>64-bit Intel</td><td>June 3, 2019</td><td>October 7, 2019</td><td>TBA</td><td>10.15.7 (19H1419)
(September 23, 2021)</td></tr>
<tr><td>macOS 11</td><td>Big Sur</td><td>20</td><td rowspan="2">64-bit Intel and ARM</td><td rowspan="2">64-bit Intel and ARM[Note 4]</td><td>June 22, 2020</td><td>November 12, 2020</td><td>TBA</td><td>11.6 (20G165)
(September 13, 2021)</td></tr>
<tr><td>macOS 12</td><td>Monterey</td><td>21</td><td>June 7, 2021</td><td>October 25, 2021</td><td>TBA</td><td>12.0 RC
(October 18, 2021)</td></tr>
</table>

Legend: Old version | Older version, still maintained | **Latest version** | Future release